AUTHENTICATING SUBLI DANCE MOVEMENTS THROUGH KEYPOINT-BASED IMAGE PROCESSING AND CNN-BASED POSTURE ANALYSIS

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Bachelor of Science in Computer Science
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APPROVAL SHEET

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R.M.R.

C.U.

ABSTRACT

This study presents a deep learning–based framework for authenticating Sublí dance movements through keypoint-based image processing and Convolutional Neural Network (CNN) posture classification, addressing the need for an objective and reliable method in evaluating traditional Filipino dance performances. MediaPipe Pose was utilized to extract skeletal keypoints from recorded Sublí dance videos, which were then converted into structured datasets representing body joint coordinates. These keypoint features were used to train a CNN model designed to classify dance postures as Correct or Not Correct based on their alignment with authentic Sublí movement standards. Experimental results demonstrate that the proposed model achieved an overall classification accuracy of 95%, with a precision of 0.95, recall of 0.96, and F1-score of 0.95 across 252 evaluated samples, indicating strong and balanced performance in posture authentication. Statistical validation using a paired t-test yielded $t(102) = -0.445$, $p = 0.657$, confirming that there was no significant difference between the system's measurement outputs and that the model produced stable and unbiased classifications. The findings indicate that integrating keypoint-based pose estimation with CNN-based posture analysis provides an effective and consistent approach for authenticating Sublí dance movements. This framework contributes to computer vision research in movement analysis while supporting Philippine cultural heritage preservation by offering a data-driven tool for dance education, performance assessment, and digital documentation of traditional practices.

Keywords - *Sublí Dance, Image Processing, Convolutional Neural Network (CNN), Human Pose Estimation, Cultural Heritage Preservation, Keypoint Detection, Computer Vision*

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DEFINITION OF TERMS

This section provides definitions for the key technical terms used throughout this study. These definitions are essential for understanding the methodologies, concepts, and components of the development of the keypoint-based image processing and (CNN) model for authenticating Subli dance movements.

Accuracy	A performance metric that measures how often the model correctly classifies dance movements. It quantifies the ratio of correct predictions to the total number of evaluated samples and, in this study, reflects how effectively the system authenticates Sublí dance movements as correct or not correct. A performance metric indicating the percentage of correct predictions made by the model.
Authentication	The process of verifying the accuracy and authenticity of a dance movement using computational analysis. It involves comparing extracted pose features against expert-defined standards to determine whether the performed Subli movement aligns with authentic patterns.
Convolutional Neural Network (CNN)	A deep learning model designed to automatically extract spatial and visual features from images. It is used in this study to classify and analyze Subli dance postures based on keypoint data, enabling the system to distinguish between correct and not correct movements.
Cultural Preservation	The process of safeguarding and sustaining traditional practices, such as dances by integrating documentation, education and modern technology. In this study, it involves using image processing and machine learning techniques to help preserve the authenticity of the Subli dance.
Dataset	A structured collection of data used for training and testing the machine

	learning model, including labeled images of dance postures, serving as the foundation for the system's learning and performance evaluation.
F1-Score	The harmonic mean of precision and recall, providing a balanced measure of a model's performance. In this study, it measures how effectively the system identifies correct and not correct Subli dance movements without bias toward either class.
Google Colaboratory (Colab)	A cloud-based platform that allows users to write and execute Python code in an online environment. It is used in this study to develop, train, and evaluate the CNN model for authenticating Subli dance movements.
Image Processing	A computational technique used to analyze and manipulate digital images to extract meaningful information. In this study, it is applied to identify and extract key posture features from Subli dance movements for model training and evaluation.
Keypoint Detection	A computer vision technique used to identify specific body joints, such as the elbows, knees, and shoulders, from images or video frames. In this study, it enables the extraction of pose coordinates necessary for analyzing and classifying Subli dance movements.
Machine Learning	A branch of artificial intelligence that enables systems to learn patterns and relationships from data to make predictions or informed decisions. It is utilized to train a model capable of distinguishing between correct and incorrect Subli dance movements based on extracted pose features.
MediaPipe	A framework developed by Google

	that enables real-time pose estimation and body landmark detection from images or videos. It is used to extract keypoints and pose coordinates of Subli dance movements for subsequent analysis and classification.
Model	A trained algorithm that processes input data to make predictions or classifications. It determines whether a Subli dance movement is authentic by analyzing extracted keypoint features and learned posture patterns.
Pose Estimation	A computer vision process that detects and maps key body joints from images or video frames to determine the subject's posture and alignment. Within the study, it is applied to analyze the movements of Subli dancers and evaluate the authenticity of their performance.
Posture Analysis	The evaluation of body position and alignment to determine the accuracy and quality of movement. It involves examining extracted pose data to assess whether Subli dance movements conform to authentic form and standards.
Precision	A performance metric that measures the proportion of correctly identified positive predictions out of all predicted positives. It reflects the model's ability to accurately classify authentic Subli dance movements without mislabeling incorrect ones.
Recall	A metric that measures the model's ability to correctly identify all actual positive instances within a dataset. It indicates how effectively the system recognizes Subli dance movements without overlooking correct postures.

Sublí Dance	A traditional folk dance from Batangas, Philippines, known for its deep spiritual significance and distinctive rhythmic movements. It embodies devotion and cultural heritage through gestures, postures, and footwork unique to the local tradition.
Training	The process of providing data to a machine learning model so it can learn patterns, relationships, and features necessary for accurate prediction. It allows the model to recognize and classify Subli dance movements based on extracted keypoint data.
Visual Studio Code (VS Code)	A code editor used for writing, testing, and debugging programs in various languages, including Python. It provides an integrated environment for developing and managing the scripts used in the implementation of the Subli dance authentication model.

CHAPTER 1

INTRODUCTION

Background of the Study

The Sublí is a devotional folk dance from Batangas performed in honor of the Holy Cross of Alitagtag. It represents a deep integration of religious devotion, cultural identity, and artistic expression within Filipino heritage. Characterized by graceful upper-body gestures from female dancers and rhythmic stick tapping by male performers, the dance embodies discipline, faith, and communal unity. Despite its cultural importance, the authenticity of Sublí performances has become increasingly difficult to preserve due to modernization, commercialization of festivals, and the absence of standardized evaluation methods for movement accuracy.

Traditional assessment of Sublí dance movements primarily depends on expert observation and oral transmission. While this approach preserves cultural context, it is inherently subjective and may vary among practitioners and instructors. Differences in interpretation, limited availability of trained experts, and the lack of formal benchmarks make consistent evaluation challenging. As a result, variations in posture, alignment, and execution may gradually deviate from authentic forms without systematic detection.

Recent advancements in computer vision and artificial intelligence offer promising solutions to these challenges. Human pose estimation techniques enable the extraction of skeletal keypoints that accurately represent body

posture and movement, while deep learning models such as Convolutional Neural Networks (CNNs) can learn discriminative patterns from these features. International studies have successfully applied pose estimation and CNN-based models to dance analysis, sports biomechanics, and motion recognition. However, these applications largely focus on contemporary or generalized movements, with limited attention given to traditional Filipino dances and authenticity verification.

In the Philippine context, the Sublí remains one of the most sacred and historically significant folk dances, yet no automated system exists to objectively authenticate its movements. Cultural festivals such as the Sublian Festival help sustain public awareness, but artistic variation and mass choreography often introduce inconsistencies in posture and form. Without a reliable and repeatable evaluation tool, ensuring the technical integrity of Sublí performances across different dancers and learning environments remains difficult.

To address this gap, this study proposes a posture authentication framework that integrates keypoint-based image processing and CNN-based posture classification for Sublí dance movements. Using pose estimation techniques, skeletal keypoints are extracted from recorded dance performances and transformed into structured numerical representations. A CNN model is then trained to classify each posture as Correct or Not Correct based on its alignment with authentic Sublí movement standards. This data-driven approach provides an objective, consistent, and scalable method for evaluating dance authenticity while respecting cultural context.

By combining intelligent systems with cultural preservation, this research contributes both technically and culturally. The proposed system supports educators, performers, and cultural institutions by offering a digital tool for movement validation, dance instruction, and heritage documentation. Ultimately, this study demonstrates how modern technology can serve as a means of safeguarding traditional Filipino practices and ensuring that the Subli dance is preserved and transmitted faithfully to future generations.

Objectives of the Study

This study aims to develop and evaluate a pose-based classification system using keypoint image processing and CNN for authenticating Sublí dance movements.

1. To develop and train a CNN model for classifying Sublí dance movements as “Correct” or “Not Correct”.

2. To implement the trained CNN model with keypoint detection on the extracted images.

3. To evaluate its performance using accuracy, precision, recall, and F1-Score.

4. To conduct a statistical analysis using paired t-test to determine significant differences between correct and not correct Sublí movements.

Significance of the Study

Preserving traditional folk dances like the Sublí is both a cultural and technological challenge in the modern era. This study addresses this by proposing an AI-driven approach that authenticates Sublí dance movements using CNN-based posture analysis and keypoint-based image processing.

Educators and Students. Teachers and learners in cultural studies, physical education, and performing arts may use the system as a digital tool to evaluate and teach Sublí movements more accurately. It encourages technology-enhanced learning that strengthens appreciation and understanding of traditional Filipino dance.

Cultural Institutions and Heritage Organizations. Agencies such as the National Commission for Culture and the Arts (NCCA) and local cultural offices can adopt the system to assess and preserve authentic Sublí performances. By ensuring that public demonstrations remain true to their traditional form, it supports cultural preservation and reinforces national identity.

Future Researchers. This study serves as a foundation for future works in heritage preservation using artificial intelligence. The researchers can expand its application to other folk dances or develop more advanced real-time authentication systems that further integrate cultural heritage with modern technology.

Scope and Limitations

This study focuses on the authentication of Subli dance movements through the use of keypoint-based image processing and Convolutional Neural Network (CNN)-based posture analysis. The data-gathering procedure was conducted on June 20, 2025, at the UPLB Carillon, located within the University

of the Philippines Los Baños campus, providing a controlled and culturally appropriate environment for capturing the performances. The primary source of data was a local Subli dancer who performed authentic Subli movements, ensuring that the dataset accurately represented traditional execution and form. To supplement the data and promote variation in poses, the researchers who participated by performing selected Subli dance movements themselves, following the same choreography, attire, and recording conditions. This approach allowed the dataset to include both authentic and non-authentic postures, providing a more balanced representation for model training and evaluation. All recordings were captured under consistent lighting and camera setups to maintain uniformity across the collected samples, supporting reliable image processing and pose estimation.

The scope of this study is limited exclusively to the Subli folk dance and does not extend to other Filipino traditional dances. The dataset used for model training and evaluation was obtained solely from video recordings collected during a single data-gathering session at the UPLB Carillon. This limited data source may restrict variations in body type, dance style, costume, lighting conditions, and background environments. Additionally, the system's performance is dependent on the quality and consistency of the captured videos, which may affect the precision of keypoint detection. Since the developed model was trained only on this specific dataset, its accuracy and generalization may decline when applied to other dancers, camera angles, movement speeds, or real-world conditions outside those controlled settings. Consequently, further validation using more diverse datasets is recommended to enhance the robustness and adaptability of the system.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review Related Literature and Studies

This chapter presents the relevant literature and studies that the researcher considered in strengthening the importance of the present study. It also presents the synthesis of related literature and studies.

Filipino Folk Dance Heritage and the Subli Tradition

Originating from Batangas Province the Sublí is a traditional devotional dance from the Philippines that embodies a profound fusion of faith, culture and identity. It is a deeply ingrained religious and cultural practice that is performed in remembrance of the Holy Cross of Alitagtag. The dance itself is unique in that women perform elegant hand gestures and circular footwork that represent supplication and reverence while men use bamboo sticks to strike rhythmic patterns. The Sublí which is a major feature of the annual Sublian Festival in July in Batangas City has transformed over time from its religious origins to become a symbol of Batangueño cultural pride. By presenting vibrant street performances, traditional clothing and coordinated group routines the festival rekindles public interest in the dance and preserves the Sublí as a form of cultural and spiritual expression (Batangas City Government n. D.). With roots in devotion, tradition and local identity the Sublí is one of the most culturally significant folk dances in the Philippines. According to DanceUs (n. d.). The Sublí was first performed as a ceremonial offering to the Holy Cross of Alitagtag

more than 300 years ago in the Batangas region. Today it is a part of town fiestas and religious events. Traditionally both male and female dancers perform the dance and their gestures convey both celebration and reverence. Women wear the traditional balintawak dress and make elegant prayer-like hand gestures while men perform rhythmic foot stomps and bamboo stick strikes while wearing the barong Tagalog and salakot. Native instruments like the octavina and bandurria provide a distinctive aural background for the dance enhancing its solemnity and spontaneity. The approach blends rigorous choreography with spontaneous moments demonstrating both self-control and emotional expression.

A symbol of Batangueño cultural pride and identity, the Sublí and the Sublian Festival are based on communal celebration and religious devotion. With its ceremonial dances, joyous processions and lively music the Sublian Festival honors the devotion to the Holy Cross of Alitagtag and reflects centuries-old customs according to Dito sa Pilipinas (2024). The annual festival in Batangas City serves as a cultural expression that upholds historical religious and communal values in addition to being a performance event. The celebrations focal point the Sublí dance highlights themes of humility, reverence and local heritage through ritualistic choreography accompanied by traditional instruments like the bandurria. The article also emphasizes how the festival helps to preserve indigenous cultural practices in a modernizing society while bringing generations of Batangueños together. For technology-based initiatives like dance authentication systems to be properly grounded in the social and performative context of the Sublí this historical and cultural background is crucial. In addition to cultural documentation and performance the preservation

of Philippine folk dance traditions like the Sublí depends on educators' dedication to passing these traditions on to the next generation. The preservation of intangible cultural heritage such as folk dance is greatly impacted by pre-service teachers' institutional involvement and personal interests according to Lobo (2023). Lobos study of 408 physical education majors revealed that students' readiness to incorporate Philippine dance traditions into their future instruction was highly influenced by both their level of personal motivation and their level of school involvement. These results indicate that in addition to community-based performances organized academic support is necessary to maintain cultural practices like the Sublí. The long-term survival of traditional dances is ensured by incorporating them into curriculum development and teacher training especially in a contemporary setting where cultural practices are increasingly at risk of marginalization.

In order to maintain Philippine folk dances practice and relevance in modern society it is crucial to comprehend the degree of interest and awareness among students. In a study published in 2024 Garcia investigated the effects of pedagogical approaches, age and location on students' interest in folk dancing. According to the results older students showed a greater awareness and appreciation for traditional dances and students from rural areas showed a higher level of interest than those from urban settings. The study also underlined how teaching strategies have a big impact on student engagement. It was discovered that interactive and tech-enhanced methods were superior to conventional lecture-based formats in terms of encouraging engagement and excitement. Additionally cultural exposure through involvement in the community school and family has been demonstrated to

enhance students' appreciation and comprehension of folk dances. These observations highlight the value of not only maintaining the technical elements of dances like the Sublí but also creating environments that foster young people's appreciation of culture via instruction and community support.

Dance Posture Analysis in Motion Studies

Recognizing posture is essential to guaranteeing the precision and expressiveness of traditional dance performances. An investigation by Wang et al. (2023) saw the proposal of a technique that used Kinect V2 vision sensors in conjunction with a skeleton-based 3D convolutional neural network (3D-CNN) to capture dance movements and recognize posture. The model was able to learn from both spatial and temporal joint features because the researchers created a dataset tailored to folk dance postures. Key skeletal joint coordinates and joint velocity data were extracted by the system and used to train the 3D-CNN classifier. With a high average recognition accuracy of 98 out of 72 percent the suggested method proved to be successful in correctly differentiating between a range of common folk dance moves. The study offers a dependable method for posture analysis in dance by concentrating on the spatial motion properties of the human skeleton and utilizing multi-frame joint features. Biomechanical research offers a scientific foundation for examining dance technique and offers important insights into how dancers perform movements and maintain posture. Krasnow et al. (2011) reviewed the literature on 89 dance biomechanical studies that were published between 1970 and 2009. The research that has already been done is categorized in the article according to topics like elevation turns, landings, rond de jambe, plié, relevé

and alignment. In all of these areas the review emphasizes that professional dancers typically exhibit more effective movement techniques than novices. It also highlights how different movement mechanics can be even among trained people suggesting that even with comparable training histories biomechanical results can vary greatly. Results from studies that used a barre revealed changes in weight distribution and different patterns of muscle activation. While the amount of biomechanical research in dance has increased the review highlights the need for additional studies that look at dynamic movement sequences especially with the use of sophisticated measurement tools. A fundamental understanding of posture and joint behavior in dance is provided by this body of work which is pertinent to systems that evaluate and analyze movement accuracy.

Frameworks that capture both spatial accuracy and qualitative aspects of performance are necessary for an accurate analysis of posture and movement in traditional dances. Laban Movement Analysis (LMA) a notation system that records elements like space effort shape and body served as the foundation for the motion analysis system that Aristidou, Stavrakis and Chrysanthou (2014) proposed for the assessment of folk dance performances. The system evaluates movement characteristics using both geometric features and stylistic parameters comparing a user's performance with that of an expert dancer using motion capture data. This method incorporates higher-level descriptors like motion intensity and fluidity in contrast to conventional systems that only concentrate on joint position or trajectory. Additionally a prototype teaching application and a 3D visualization interface were presented in the study giving students access to both quantitative and visual feedback. In

comparison to traditional distance-based metrics their analysis showed that LMA features offered more distinct differentiations between novice and expert performances. This study provides a strong basis for assessing posture in culturally specific dance forms by highlighting the significance of both biomechanical and expressive characteristics in examining traditional dance movements. Dance posture changes over time are a reflection of changing performance styles and aesthetic standards. Daprati, Iosa and Haggard (2009) used segmental angles to measure changes in body alignment in archival photos of dancers in the same classical ballet positions from the 1940s to the 2000s. Compared to their predecessors modern dancers displayed a more vertical posture according to their analysis which showed a distinct temporal trend. Participants in an evaluation by the authors rated simplified stick-figure representations of the extracted postures in order to determine whether these changes had aesthetic significance. Recent vertically aligned postures were deemed more aesthetically pleasing according to the results. Using both objective kinematic features and subjective perception the study shows how biomechanical posture can be assessed and valued. It draws attention to the potential of posture analysis for both movement evaluation and comprehending the aesthetic development of dance performances over time.

For dance instruction and movement correction to be effective real-time posture analysis is crucial. Wang (2021) suggested a spatial data partitioning-based posture analysis technique that tracks and assesses dancers movements using Kinect bone tracking technology. By splitting the spatial region into several sections and capturing skeletal joint data in real time the system allows for targeted analysis of particular body segments while motion is

being executed. Through the comparison of dynamic joint positions with reference data this method enables the effective detection of posture deviations. The study highlights how crucial real-time feedback is to improving dance learning outcomes and lowering practice errors. The application of the technique seeks to enhance dance instruction by enabling students to see problems with posture alignment and make quick corrections. The study demonstrates how motion capture and spatial data analysis can greatly enhance accuracy in dance posture evaluation. Dancers' motor skill development can be seen in patterns of movement responses in various learning environments. Castañer et al. (2009) examined how pupils reacted to kinesthetic metaphoric and descriptive instructions in tasks involving mime and creative dance. Stable motor patterns emerged among learners according to the study which used an observational methodology in conjunction with temporal pattern (T-pattern) detection. The findings revealed that participants produced coordinated locomotion sequences and preferred support and axial stability. Learners consistently internalized motor sequences as evidenced by their replication of structural elements of movement included in the instructions even though the tasks were creative. The study provides important insights into how posture and movement change in response to different teaching philosophies by highlighting the ways in which specific postural and sequential motor traits appear early in the dance learning process.

Keypoint Detection in Human Pose Estimation

Several applications in motion capture action recognition and gesture analysis are made possible by human pose estimation (HPE) which has

emerged as a fundamental task in computer vision. Lan et al. (2022) reviewed significant developments in 2D and 3D pose estimation in a thorough survey of vision-based deep learning-based human pose estimation techniques carried out in 2022. In the paper popular architectures like OpenPose HRNet and DensePose are compared and methods are divided into top-down and bottom-up frameworks. Additionally it highlights the significance of essential elements such as heatmap regression multi-stage refinement and backbone networks in raising prediction accuracy. In order to benchmark pose estimation models a large amount of the survey is devoted to datasets such as COCO MPII and Human3. 6M as well as evaluation metrics like Mean Per Joint Position Error (MPJPE) Area Under the Curve (AUC) and Percentage of Correct Keypoints (PCK). Lan et al. emphasize even more the enduring difficulties in the field like generalization across various scenes, complicated multi-person interactions and occlusion. Their survey is a useful resource for choosing and modifying pose estimation techniques appropriate for jobs like examining and verifying traditional dance moves where accuracy in keypoint identification is essential. Both culturally-adapted data and real-time processing are needed for the recognition of traditional dance movements. In order to analyze Korean traditional dance movements Kim (2025) proposed a recognition system that makes use of BlazePose, a vision-based deep learning model that can extract 33 skeletal keypoints. The integration of a metadata-enriched dataset which contained positional temporal and contextual information about the dancer and environment improved pose estimation performance in the study. In order to guarantee culturally appropriate interpretation and enhance the systems recognition capabilities this metadata was utilized in conjunction with the pose

data. To better adapt general pose estimation models for conventional movement styles the study highlights the significance of including domain-specific knowledge in training data.

Because they allow for more flexible modeling of spatial relationships between keypoints, transformer architectures have demonstrated promising advances in the field of human pose estimation. Li et al. (2021). presented TokenPose a Transformer-based model that learns explicit keypoint token representations to estimate human poses. This architecture encodes each anatomical keypoint as a learnable token that uses self-attention mechanisms to interact with other keypoint tokens and image patch tokens. This structure enables the model to concurrently capture anatomical constraints and visual appearance cues. When compared to models based on convolutional neural networks (CNNs) TokenPose produced competitive results when tested on industry-standard benchmarks such as COCO and MPII. For instance compared to earlier Transformer and CNN baselines the TokenPose-L model used fewer parameters and required less computation to achieve 75. 8 Average Precision (AP) on the COCO validation set. The study shows that an efficient and successful substitute for conventional heatmap-based pose estimation techniques is to encode human keypoints as explicit tokens inside a Transformer framework. Conventional methods of keypoint detection in human pose estimation usually handle each keypoint separately frequently with the use of regression-based outputs or heatmaps. McNally et al. (2022) suggest a change in this approach by representing complete human poses as well as individual keypoints as objects. This object-centric formulation eliminates the need for distinct association steps that are typical of bottom-up approaches and

enables unified detection and grouping within the same pipeline. The center of a pose and its associated keypoints can be predicted by the network using a center-offset representation which treats the problem similarly to object detection. The suggested model showed competitive performance when tested on the COCO and CrowdPose datasets providing enhanced simplicity and interpretability. The authors stress that their method provides a fresh viewpoint on pose estimation by considering keypoints as objects which facilitates more coherent structure and collaborative reasoning in crowded or complex scenes.

The goal of bottom-up methods for human pose estimation is to locate each important point in a picture before linking it to specific human cases. Geng et al. (2021). By dividing the representation of each keypoint through a separate regression branch the novel framework known as Disentangled Keypoint Regression (DEKR) which was proposed in 2021 enhances keypoint localization. In order to independently learn each keypoint within its own branch of the network the method disentangles the learning process and uses adaptive convolutions to extract region-specific features. This design improves localization accuracy and reduces interference between various keypoints particularly in crowded or complex scenes. Tests on the COCO and CrowdPose datasets showed that DEKR performed better than the most advanced bottom-up methods in terms of robustness and average precision.

As an efficient and effective substitute for traditional heatmap-based keypoint detection the authors emphasize that their regression-based approach achieves higher accuracy with less post-processing.Precise keypoint localization in human pose estimation depends on maintaining high-resolution

representations across the network. Sun et al. (2019). In contrast to conventional architectures the High-Resolution Network (HRNet) was introduced in 2019. It preserves high-resolution feature maps throughout all layers instead of recovering resolution after downsampling. High- and low-resolution features can effectively exchange information thanks to the HRNet architecture's parallel subnetworks that process various resolutions and repeatedly perform multiscale fusion. Because spatial precision is maintained this design results in more accurate keypoint detection. In tests using common benchmarks like COCO and MPII HRNet outperformed the state-of-the-art techniques available at the time obtaining greater accuracy in pose estimation tasks involving one or more people. The study demonstrates how well multiscale feature fusion and high-resolution representation work to enhance human pose estimation systems performance.

Convolutional Neural Networks (CNN) in Image-Based Analysis

Recently convolutional neural networks (CNNs) have been used in image-based posture analysis to assess how accurately people move in performance environments. In a study published in 2025 Lian integrated a deep convolutional embedding attention mechanism into an image recognition framework to optimize dance movement recognition for theatrical applications. The system's main goal was to recognize common dance steps such as square steps, time steps and center of gravity transitions with an overall accuracy (OA) of 98–65% and recognition accuracy of over 80% for each. In terms of parameters and floating-point operations (FLOPs) the suggested model maintained computational efficiency while providing a 12% improvement over

baseline approaches. This research validates CNNs ability to accurately analyze expressive human movement in addition to their technical capabilities. Accurate classification in vision-based systems depends on an understanding of how image compression impacts the performance of deep learning models. Hamano et al. (2023) used a Vision Transformer (ViT) model within an encryption-then-compression (EtC) framework to examine how JPEG compression affected the accuracy of image classification. Using the CIFAR-10 and ImageNet datasets the study assessed classification performance across a range of JPEG quality factors. The ViT model achieved over 98 percent accuracy on CIFAR-10 even at a quality factor of 85 according to the results proving that classification accuracy can withstand moderate compression. The study also found that in compressed and encrypted environments ViT performed better than models based on convolutional neural networks (like ResNet and MobileNet). Additionally because ViT maintained high classification performance without requiring image decryption the study demonstrated the benefits of using it in situations where image privacy is an issue. These results offer important new information for applications involving compressed visual data where image-based classification systems need to strike a balance between accuracy and data efficiency.

Dance postures in videos must be recognized using models that can handle both temporal and spatial movement variations. To identify poses from dancing images in an Internet of Things (IoT) setting Zheng and Yuan (2023) proposed the Dual Convolutional Neural Network Algorithm (DCNNA). The technique records appearance and motion cues by fusing optical flow features with frame-by-frame spatial information. The model is better able to identify

intricate dance-related body postures thanks to DCNNA which structures the network to process spatial and temporal inputs independently before fusing them. In the F1 score the authors found that their method performed better than traditional models like Inception V3 and 3D-CNN improving it by roughly 11% and 6% respectively. The study illustrates the benefit of employing convolutional neural networks to integrate multiple feature representations for accurate and effective posture recognition in performance contexts. It is a challenging visual task to identify subtle variations in human posture especially in fields like dance, yoga and sports. Bera et al. (2023) carried out a benchmark analysis on convolutional neural networks (CNNs) for fine-grained posture recognition. In addition to two other datasets for yoga and sports the study presented the Dance-12 dataset which includes 12 classes of traditional Indian dance poses. Several CNN-based backbone models including MobileNetV2 ResNet50 and EfficientNetB0 were tested both with and without a patch-based attention (PbA) module. To increase classification accuracy the PbA module was created to highlight spatially relevant image regions. Visual attention mechanisms are effective in fine-grained pose classification as evidenced by experimental results showing that integrating PbA with CNN architectures significantly improved performance across all datasets including Dance-12. For CNN-based systems tasked with differentiating intricate human movements in culturally specific dance forms this study offers a solid basis.

Capturing both temporal consistency and spatial accuracy is necessary for assessing the quality of dance movements. Qin and Meng (2025) suggested evaluating dance motion quality from video input by employing spatiotemporal convolutional neural networks (ST-CNNs). From video clips of dance

performances the model extracts features in both space and time then maps those features to a numerical score that represents the movements perceived quality. The model was trained and validated using a dataset of annotated dance clips with expert assessments acting as ground truth. The findings demonstrated that the ST-CNN model could successfully capture dance characteristics like expressiveness control and fluency since the predicted scores closely matched expert ratings. The study demonstrates how CNN-based models can produce unbiased evaluations of dance quality providing useful resources for dance education and performance analysis. In classical dance gesture recognition is crucial for maintaining and interpreting culturally significant nonverbal communication. By developing a specialized video benchmark dataset and using deep learning techniques for real-time gesture classification Parthasarathy and Palanichamy (2023) introduced a novel method for identifying symbolic hand gestures in Indian classical dance. To represent the variability found in the real world the dataset includes a large variety of hand gestures captured under various circumstances. For feature extraction and classification the study used convolutional neural networks (CNNs) and metrics like accuracy and processing time were used to assess the models performance. The suggested system proved to be suitable for interactive applications and educational settings by exhibiting high recognition accuracy while retaining real-time responsiveness. In addition to helping with the automatic classification of expressive dance gestures the authors highlighted the importance of their dataset and recognition model in supporting the digital preservation of Indian cultural practices.

Machine Learning Applications in Cultural Heritage and Performance Arts

Incorporating digital technologies into cultural preservation is a way to resist the marginalization of indigenous narratives as well as a matter of innovation. Digital humanities are a cultural defense mechanism according to De Leon (2025) giving Filipinos the opportunity to reframe and reclaim their cultural identity using easily accessible digital tools. This viewpoint is particularly pertinent in the context of traditional performance arts like the Sublí folk dance where heritage preservation and representation are closely related. Researchers are enhancing movement recognition accuracy and furthering the larger goal of cultural empowerment and continuity by incorporating machine learning and image-based analysis into these cultural expressions. Through the use of these technological tools customs can be actively preserved and it is ensured that indigenous art forms are not only preserved but also actively engaged with respect and passed down to future generations. With an increasing focus on strategic integration in both tangible and intangible heritage domains the use of artificial intelligence (AI) in cultural heritage preservation is accelerating. Harisanty et al. (2024) carried out a bibliometric study that delineated the terrain of AI-powered heritage conservation pinpointing five major technological clusters: software development virtual exhibitions conservation evaluation exhibition visualization and metadata and database administration. These clusters show how artificial intelligence (AI) tools like machine learning are being used to reimagine heritage experiences using robotics drones, virtual and augmented reality (VR/AR) and 3D modeling in addition to digitizing and organizing cultural assets. The report highlights how cultural organizations must carefully consider and implement AI advancements

to improve sustainability accessibility and curation. These technological advancements facilitate the application of intelligent systems for the analysis, preservation and presentation of movement-based heritage in traditional performance arts like the Filipino Sublí dance. The study reaffirms machine learning's potential as a breakthrough tool for the long-term preservation and appreciation of cultural traditions by bridging the technical and cultural domains.

By combining the documentation and dissemination of cultural heritage with information and communication technologies (ICT) digital humanities (DH) are becoming more and more important in cultural preservation. According to Khan et al. (2016) DH gathers, organizes and disseminates cultural knowledge using a variety of digital resources such as print audio video graphics and other media formats. The chapter discusses several DH activities including data mining metadata standardization, repository development and digital curation all of which are used to preserve aspects of cultural identity like craft designs, folklore rituals and customs. It highlights how crucial accessibility and sustainability are to cultural preservation initiatives and looks at how academic and archival institutions can help preserve collections of digital heritage. To guarantee the preservation of cultural materials for future generations the authors also discuss the difficulties associated with digitization and promote the use of collaborative digital platforms. Because digital repositories and structured metadata can facilitate both conservation and analysis this framework is especially pertinent to computational efforts aimed at maintaining performance-based traditions. Traditional performing arts like folk dance can now be systematically recorded, analyzed and visualized thanks to the incorporation of digital technologies into cultural heritage preservation. Kico et

al. (2018) examined the methods and technologies currently in use for digitizing and visualizing folk dances highlighting their importance in protecting intangible cultural heritage. The study examines various motion capture systems including mechanical inertial and optical ones that are used to capture dancers movements in high definition. These platforms facilitate the development of interactive visualizations and digital archives that support dance performance evaluation and instruction. Additionally the authors looked at visualization techniques like virtual reality and 3D animation which give students immersive experiences and real-time feedback. Digital platforms can help preserve and transmit folk dance traditions as the paper illustrates by addressing the technical cultural and educational aspects of digitizing dance.

The analysis and preservation of dance as a type of intangible cultural heritage (ICH) can be facilitated by machine learning (ML). Rallis et al. (2020) examined recent advancements in the use of machine learning techniques for dance analysis with a focus on the categorization and identification of dance movements, gestures and styles. The review describes the particular algorithms used such as convolutional neural networks (CNNs) recurrent neural networks (RNNs) and hybrid CNN-RNN architectures and classifies methods according to the input modality, such as motion capture video or multimodal data. The purpose of these models is to identify characteristics such as rhythm spatial placement and choreography. In their discussion of supervised and unsupervised learning strategies the authors point out that the availability of annotated datasets has led to the current dominance of supervised methods. In order to properly depict the dynamic nature of dance the paper also emphasizes the significance of temporal modeling and multimodal fusion.

Although machine learning provides scalable tools for recording and analyzing traditional dance the review comes to the conclusion that more work is required in the areas of evaluation metrics dataset development and culturally sensitive representations. Artificial intelligence (AI) is being used more and more in the fields of cultural heritage education presentation and preservation. Through a thorough bibliometric analysis of 3253 publications released between 2019 and 2023 Gîrbacia (2024) investigated AI-related research in cultural heritage. Numerous technological trends were identified by the study such as the application of computer vision, machine learning, 3D reconstruction, virtual and augmented reality and more. The main uses of these technologies are in the fields of interaction design, cultural data analysis, virtual exhibitions and heritage digitization. The analysis also showed that user-centered applications and interdisciplinary partnerships are becoming more and more popular. Through the mapping of keywords author networks and publication distributions the study demonstrated how AI uses creative data-driven methods to support cultural heritage. The results offer a comprehensive summary of the ways in which machine learning and associated technologies are advancing the conservation and dissemination of cultural heritage.

Human Action Recognition and Classification in Computer Vision

Architectures that process spatial and temporal information simultaneously are useful for analyzing dance movements in complex environments. A double-stream convolutional neural network specialized for dance action recognition is presented by Jin (2022). One stream uses optical flow inputs to process temporal information while the other stream processes

spatial data from video frames. The study emphasizes that even when the number of individuals in the scene increases this dual-stream structure maintains efficient computation and improves recognition performance. In contrast to other algorithms which scale linearly with crowd size the authors observe that running time increases very slowly with the number of subjects. The framework has practical value for automated dance motion recognition systems because it can be applied to fields like dance teaching, dance video retrieval and the protection of art and cultural heritage. Analyzing how motion is processed and remembered requires knowing whether human action recognition depends more on dynamic movement or static posture. Vicary et al. (2014) used two experiments with point-light displays of dance-like movements to examine this question. In the first experiment participants had to decide whether two consecutive clips were the same or different after watching them. According to the results identifying complete movement sequences were much more accurate than identifying single static postures. The researchers tested the effects of two different types of interfering stimuli on memory in the second experiment: one based on motion (a distinct movement sequence) and the other based on form (a static frame). They discovered that dynamic interference more severely hampered movement recognition than static interference indicating that motion rather than form is the primary basis for human action memory. For human action recognition the study finds that dynamic cues predominate in working memory highlighting the significance of constant movement in the encoding and retrieval of actions.

Cai et al. (2023). suggested utilizing a sequence-to-sequence (Seq2Seq) network to generate dances based on music. A multi-scale fusion

high-resolution network is used to extract dance and music features at the start of the system. A gated recurrent unit (GRU)-based encoder is then used to encode these features and a GRU-based decoder is used to decode them into a series of dance pose representations. In order to produce dance that is both structurally coherent with conventional movement patterns and rhythmically consistent with the music the model is trained. Visual analysis and sequence similarity comparisons with real dance data were used by the authors to assess their approach. Their method maintained fluidity and realism while successfully synchronizing music and generated dance according to the results. By showing how sequential modeling can be used to automatically synthesize culturally contextualized dance movements this work advances the field of action generation and recognition. Using artificial intelligence and augmented reality Jayanthi and Maheswari (2024) demonstrated a technique for 3D reconstruction and pose recognition in Bharatanatyam dance. The study created skeleton-based representations and extracted body landmarks from photos of dance poses using MediaPipe. A model called InceptionResNetV2 was trained using these skeletonized outputs and it used the skeletal data to classify various Karana poses. To increase training efficiency the researchers used image preprocessing methods like data augmentation and skeletonization. Their approach demonstrated that skeletonized input improves classification performance by achieving high accuracy in recognizing Karana poses from both synthetic and real images. The authors also created a 3D visualization of the classified poses using an augmented reality interface built on Unity.

For the preservation and analysis of culturally significant nonverbal communication gesture recognition in classical dance is crucial. By developing a specialized video benchmark dataset and using deep learning techniques for real-time gesture classification Parthasarathy and Palanichamy (2023) introduced a novel method for identifying symbolic hand gestures in Indian classical dance. The dataset includes a large variety of hand gestures that were captured in various settings to represent the variability found in the real world. Convolutional neural networks (CNNs) were used in the study to extract and classify features and indicators like processing time and accuracy were used to assess the models performance. The suggested systems high recognition accuracy and real-time responsiveness validated its applicability for interactive applications and learning environments. In order to support the automatic classification of expressive dance gestures and aid in the digital preservation of Indian cultural practices the authors underlined the importance of their dataset and recognition model. Models capable of extracting and interpreting intricate visual motion features are necessary for the accurate recognition of body movements in dance videos. Qiao and Shen (2021) combined image transformation techniques with neural network-based classification to propose a human action recognition method specifically tailored for dance video analysis. In order to improve the motion characteristics of the dance sequences, the system first extracts body posture features from video frames and then applies transformation algorithms. An input neural network model trained to categorize particular dance moves receives these processed features. The method seeks to increase movement recognition effectiveness and precision in performance video settings. Because dance movements frequently involve

complex body configurations and quick transitions the author underlined the significance of modifying recognition models to account for this visual complexity. The study shows how action recognition systems in dance applications can function better when image processing and machine learning are combined.

In a recent study, Xiang Ju (2025) proposed a fusion-based deep learning framework for dance pose estimation that combines High-Resolution Network (HR-Net) for high-resolution feature extraction with ResNet-152 as the classifier, connected through a fusion layer to capture both local and global pose patterns. The paper also tackled class imbalance in dance datasets by applying SMOTE during preprocessing, and reported strong classification metrics (accuracy ≈ 0.987), demonstrating improved discriminative power for dance pose recognition. While the approach substantially refines static pose representation and feature extraction, the authors noted limitations in modeling dynamic transitions between poses and raised concerns about potential overfitting when synthetic sampling is used. Ju's work reinforces the value of advanced deep architectures and global–local feature fusion for pose-based movement analysis and offers methodological insights that inform the present study's CNN-based posture authentication of Sublí dance movements.

Literature Matrix

Table 1. Literature Matrix

Author (s)	Title	Year	Method/Algorithm	Key Findings
De Leon	Digital Humanities as Cultural Resistance	2025	Theoretical analysis of DH + AI in heritage	Advocated digital tools for reclaiming cultural identity, especially in traditional arts like Subli.

Lian	Optimization of Image Recognition Technology for Dance Theatre Creation and Evaluation of Its Effectiveness	2025	CNN with attention-based embedding	Achieved 98.65% accuracy; improved performance in classifying movement sequences with reduced FLOPs.
Kim	Real-Time Recognition of Korean Traditional Dance Movements Using BlazePose and Metadata-Enhanced Framework	2024	BlazePose with contextual metadata	Culturally adapted dataset improved recognition performance for traditional dances.
Garcia	Interest and Awareness of Learners on Philippine Folk Dance	2024	Quantitative survey in rural/urban schools	Found rural students more engaged; tech-enhanced teaching improved learning and appreciation of folk dance.
Dito Sa Pilipinas	Discovering Subli and Sublian Festival's Cultural Mystique	2024	Cultural Journalism	Emphasized Subli's symbolic value in Batangueño identity; noted modern drift from authentic movements.
Girbarcia	Cultural Heritage Preservation in the Digital Age: A Bibliometric Analysis	2024	Bibliometric review (3, 253 AI studies)	Identified AI's global rise in heritage use: pose analysis, 3D modeling, and intelligent classification for cultural preservation.
Wang et al.	Dance Posture Analysis in Motion Studies Using Kinect and 3D-CNN	2023	3D-CNN + Kinect bone tracking	Achieved 98.72% accuracy; effective in posture-based folk dance classification using joint motion data.
Parthasarathy & Palanichamy	Gesture Recognition in Indian Classical Dance Using CNN	2023	CNN on symbolic Bharatanatyam Gestures	High real-time accuracy in recognizing complex cultural hand gestures; supports interactive learning tools.
Lan et al.	A Comprehensive Review of Markerless Motion Capture Systems in Sports	2022	Survey of OpenPose, HRNet, DensePose	MediaPipe and OpenPose rated highly in keypoint accuracy, important for traditional dance verification.
Rallis et al.	Machine Learning for Dance Movement Recognition: A Review	2020	Review of CNN, RNN, hybrid models	CNN ideal for posture; RNN for sequences; emphasized multimodal inputs and cultural dataset importance.

Synthesis of the Study

While numerous studies have applied human pose estimation, convolutional neural networks, and deep learning techniques to dance analysis, human action recognition, and movement classification, most existing works

focus on contemporary dance, sports motion, or generic activity recognition. Studies that examine traditional dances primarily emphasize performance recognition, choreography generation, or aesthetic evaluation rather than movement authenticity verification.

Furthermore, international research on traditional dance authentication largely centers on Korean, Indian, and Western dance forms, with limited focus on Filipino folk dances, particularly the Sublí. Existing systems also rely heavily on temporal or multi-sequence modeling, which may not be suitable for posture-based cultural authentication where form accuracy and alignment are critical.

At present, there is no known automated system that integrates keypoint-based image processing and CNN-based posture classification specifically designed to authenticate Sublí dance movements. This study addresses this gap by proposing an objective, posture-focused authentication framework tailored to the cultural and movement characteristics of the Sublí dance, contributing both to intelligent systems research and Philippine cultural heritage preservation.

Research on keypoint detection frameworks, such as MediaPipe, OpenPose, and HRNet (Lan et al., 2022; Kim, 2024), further highlights the importance of precise skeletal landmark extraction in motion analysis. These tools enable accurate representation of joint positions and body alignment, which are essential for evaluating traditional dance postures. However, most studies utilize these frameworks for general action recognition or performance analysis rather than authenticity verification.

From a cultural perspective, studies on traditional and classical dances emphasize the importance of preserving movement integrity and symbolic meaning (Parthasarathy & Palanichamy, 2023; De Leon, 2025). While international efforts demonstrate how artificial intelligence can support cultural heritage preservation, local studies reveal that Philippine folk dances, including the Sublí, remain vulnerable to inconsistency due to subjective assessment and lack of standardized evaluation tools (Garcia, 2024; Dito sa Pilipinas, 2024).

Collectively, the literature indicates a clear need for an objective, posture-based authentication system that balances technical accuracy with cultural sensitivity. Unlike prior studies that prioritize sequence modeling or expressive performance scoring, this research focuses on postural correctness and alignment, which are fundamental to maintaining the authenticity of the Sublí dance. By integrating keypoint-based image processing with CNN-based posture analysis, the present study extends existing methodologies into a culturally specific context, offering a practical solution for dance education, documentation, and heritage preservation.

Conceptual Framework of the Study

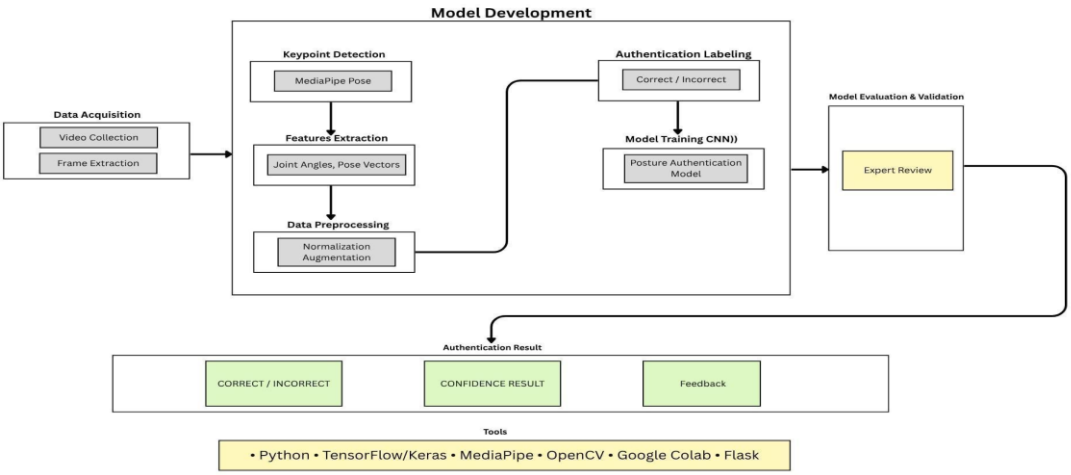


Figure 1. Conceptual Framework

The conceptual framework of this study presents a step-by-step process that aims to preserve the authenticity of the Subli folk dance by utilizing modern technologies such as image processing and machine learning. The primary objective is to develop an intelligent system capable of identifying whether a dancer's movements are authentic or not. The process begins with data acquisition, where video recordings of Sulbi dance performances are collected and used as input for the system. These videos allow the system to analyze and interpret the unique movement patterns of the dance. The next step is feature extraction, which involves using tools like Mediapipe or OpenPose to detect key body joints such as the elbows, shoulders, and knees. From these detected keypoints, numerical values such as joint angles and posture vectors are computed. These features serve as the system's understanding of how Subli movements should appear.

Once features are extracted, the data is divided into a training set and a testing set. The training set is used to teach the machine learning model how to recognize authentic Sublí movements, while the testing set evaluates how well the model can identify new, unseen inputs. For model development, a Convolutional Neural Network (CNN) is trained to classify whether a movement is "CORRECT" (authentic) or "Not Correct" (not authentic). The model's performance is measured using metrics such as accuracy, precision, recall, and a confusion matrix to ensure the system makes reliable predictions.

To further ensure accuracy and cultural relevance, the system undergoes model evaluation through user testing and expert validation. Ordinary users try the system to test its usability, while Sublí dance experts

confirm whether the system's analysis matches traditional dance standards. Once the model passes these evaluations, it proceeds to model activation, where it is deployed for public use. When a user uploads a dance video or image, the system analyzes it and generates a result indicating whether the performance is authentic or not.

This entire framework is built using tools such as Python, Google Colab, OpenCV, Mediapipe/OpenPose, and Visual Studio Code. By integrating technology with cultural preservation, this system not only promotes innovation but also contributes to safeguarding Filipino heritage. Ultimately, the framework ensures that future generations can continue to experience and perform the Sublí dance in its most authentic form.

CHAPTER 3

METHODOLOGY

Study Area

This research focuses on the preservation of Filipino cultural heritage through the application of image processing and machine learning to authenticate traditional folk-dance movements, specifically the Sublí dance. The filming was conducted at the University of the Philippines Los Baños (UPLB) Carillon. No fieldwork was conducted in formal institutions, museums, or cultural centers. Instead, the study relied on a community-based approach to data gathering and consultation.

The primary source of information and reference for this study was a local Sublí dancer from Bayog, Los Baños, Laguna. The dancer, who has personal experience and knowledge of the traditional Sublí dance, provided valuable insights into the correct body movements, steps, and posture associated with authentic performances. Through this consultation, the researchers were able to gather reliable data and cultural context, which served as the basis for model training and validation.

This approach emphasizes the importance of local knowledge in cultural preservation, especially in cases where traditions are passed down through generations and preserved within communities rather than formal archives.

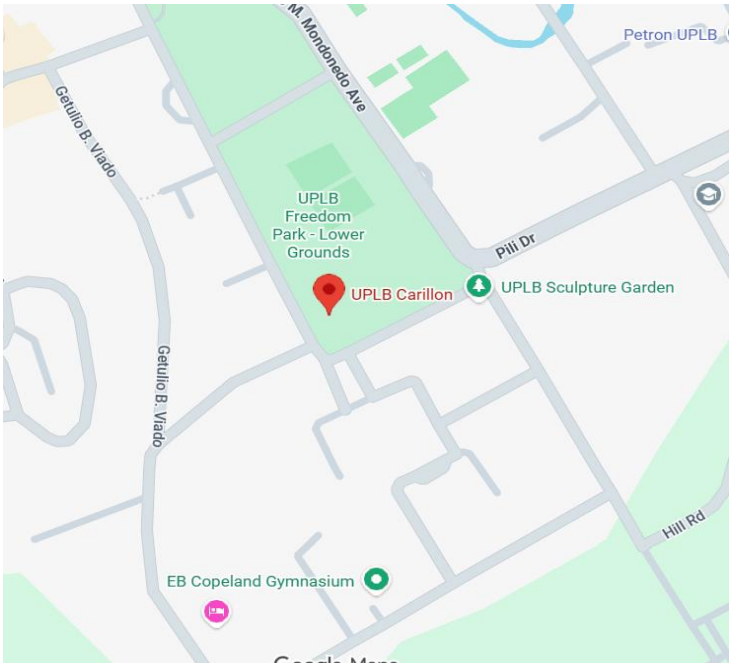


Figure 2. Location of UPLB Carillon

Materials

The researchers used the following sets of materials and sources to conduct thorough research.

Software

Here is a list of the key software tools that will be utilized in this study, including the integrated development environment, framework, packages and libraries, and the overall development of the research

Google Colaboratory (Colab)

Google Colaboratory, or Colab, is a cloud-based platform that allows users to run Jupyter Notebooks in a web browser. It provides access to powerful computing resources, including GPUs, making it ideal for machine learning tasks. In this study, Colab served as the main environment for preprocessing

the dance video data and training the convolutional neural network (CNN). It was used to process keypoint data from images, split the dataset, train the model, and visualize performance metrics such as accuracy and loss.

Visual Studio Code

Visual Studio Code is a free and open-source code editor developed by Microsoft. It supports multiple programming languages and is equipped with features that enhance code management and debugging. In this study, VS Code was used to organize and develop scripts for model integration, preprocessing automation, and managing system components outside of Colab , such as for deploying or integrating the posture analysis model into a user-facing system.

Python Libraries

A high-level, interpreted programming language known for its simplicity and readability. It offers a wide range of libraries for various applications. In this study, specific Python libraries are used to aid in tasks ranging from data preprocessing and analysis to model development and evaluation. Here are the related libraries used in the development of the study.

Mediapipe

Mediapipe is an open-source framework developed by Google that enables real-time perception pipelines for computer vision tasks. In this study, Mediapipe was used specifically for pose estimation, which involves detecting human body landmarks or keypoints such as joints (e.g., elbows, knees, shoulders) from video frames. This was a critical step in the research, as the

authentication of Sublí dance movements relied on the accurate detection and tracking of these body positions throughout the dance performance. Mediapipe provided high-fidelity results with low computational cost, making it suitable for both real-time processing and batch analysis. The keypoints extracted using Mediapipe served as the primary input features for the CNN-based posture classification model.

Pandas

Pandas is a powerful data manipulation and analysis library in Python, well-suited for handling structured data. In the context of this study, Pandas was used to manage and organize metadata related to the dataset. This included information such as file names, authenticity labels (e.g., authentic vs. non-authentic), timestamps, and the dancer's identity (if applicable). Pandas was instrumental in creating clean and well-structured data tables, filtering relevant information, and preparing the dataset for further analysis. It enabled easy merging of datasets, removal of corrupted or irrelevant data entries, and reorganization of information to match the format needed for training the machine learning model.

NumPy

NumPy is a Python library that supports large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays. The researchers utilized NumPy for efficient numerical computations, handling image data as arrays, and performing complex mathematical calculations necessary for preprocessing (e.g., normalization) and analyzing image features.

Matplotlib

Matplotlib is a Python library for creating various types of visualizations, including static, animated, and interactive plots. In this study, it was used to display sample images, visualize feature distributions, and evaluate model performance visually (e.g., plotting accuracy curves, confusion matrices). These plots allowed for clearer interpretation of results and effectively communicated key findings, aiding in data-driven decision-making throughout the analysis process.

Scikit-learn (sklearn)

Scikit-learn is a widely-used machine learning library that provides tools for data preprocessing, classification, regression, clustering, and model evaluation. In this study, Scikit-learn was essential for evaluating the performance of the CNN-based posture classification model. It was used to calculate important metrics such as accuracy, precision, recall, and to generate confusion matrices that showed how well the model distinguished between authentic and non-authentic dance postures. Additionally, Scikit-learn was used for splitting the dataset into training and testing sets and for applying standardization techniques to normalize data inputs before feeding them into the model.

TensorFlow/Keras

TensorFlow is a powerful open-source platform for building and deploying machine learning models, while Keras is its high-level API that simplifies the process of defining, training, and testing neural networks. In this

study, TensorFlow/Keras was used to develop the Convolutional Neural Network (CNN) that served as the core of the posture authentication system. These tools enabled the researchers to design the model architecture, configure layers, apply activation functions, and optimize the learning process using different hyperparameters. The CNN was trained to recognize patterns in keypoint-based data and classify whether a given dance posture matched the authentic Sublí style. TensorFlow also allows model tuning, performance monitoring, and saving trained models for future deployment.

Gradio

The Gradio framework was utilized to develop the system's interface due to its efficiency, flexibility, and seamless integration with machine learning models. Gradio provides an intuitive platform for deploying deep learning systems without requiring extensive web development, enabling real-time interaction between users and the trained CNN model. Through its built-in components for image and video input, the interface allows users to upload Sublí dance performances, visualize detected poses, and receive immediate classification feedback.

Hardware

The computer serves as the fundamental piece of hardware, providing the necessary computational resources for the development and evaluation of the authentication model. In this study, the researchers utilized their own wide range of tasks involved in the refinement and development of the model.

Table 2. Laptop Specification

Laptop Specification	
PROCESSOR	Intel(R) Pentium(R) Silver N6000 @ 1.10GHz 1.11 GHz
RAM	8.0 Gigabytes
GPU	Intel® UHD Graphics 600
SOLID-STATE DRIVE(SSD)	238 Gigabytes

Research Design

The study utilized a descriptive–experimental research design. The descriptive component focused on presenting and explaining the process of developing the authentication system for Sublí dance movements using keypoint-based image processing and Convolutional Neural Networks (CNNs). It described the steps undertaken in data collection, keypoint extraction, model development, and system implementation.

Meanwhile, the experimental component involved testing and evaluating the performance of the developed CNN model under controlled conditions. The system was trained and tested using collected Sublí dance data, and its performance was assessed through standard evaluation metrics such as accuracy, precision, recall, and F1-score. This approach allowed the researchers to both describe the development process and experimentally validate the effectiveness of the system in authenticating Sublí dance movements.

Model Development

The model development process for authenticating Sublí dance movements consists of six key stages: data acquisition, keypoint detection, feature extraction, model training, model evaluation, and system output generation. The process begins with the data acquisition phase, where videos of authentic Sublí dance performances are collected using accessible recording devices such as smartphones. These video samples serve as the foundation for training and testing the system. In the keypoint detection stage, body joints such as the head, shoulders, elbows, hips, and knees are identified using human pose estimation tools like Mediapipe or OpenPose. This converts the dancer's movements into a structured set of body coordinates. Next, the feature extraction stage transforms the detected keypoints into quantifiable features. This includes calculating joint angles, generating pose vectors that capture the direction and distance between joints, and creating skeleton images which visually represent posture. These extracted features are essential for training the machine learning model. The data is then split into training and validation sets to prepare for model development. During the model training stage, a Convolutional Neural Network (CNN) is trained to classify whether a given dance posture matches the authentic Sublí pattern based on the extracted pose features. After training, the model undergoes evaluation using standard metrics such as accuracy, precision, recall, F1-score, and a confusion matrix. These metrics help assess the model's reliability and ability to generalize to new data. The system output then provides an authentication result. It identifies whether the detected posture is authentic or not authentic, and also offers feedback on

specific posture errors, such as incorrect arm position or misaligned foot placement.

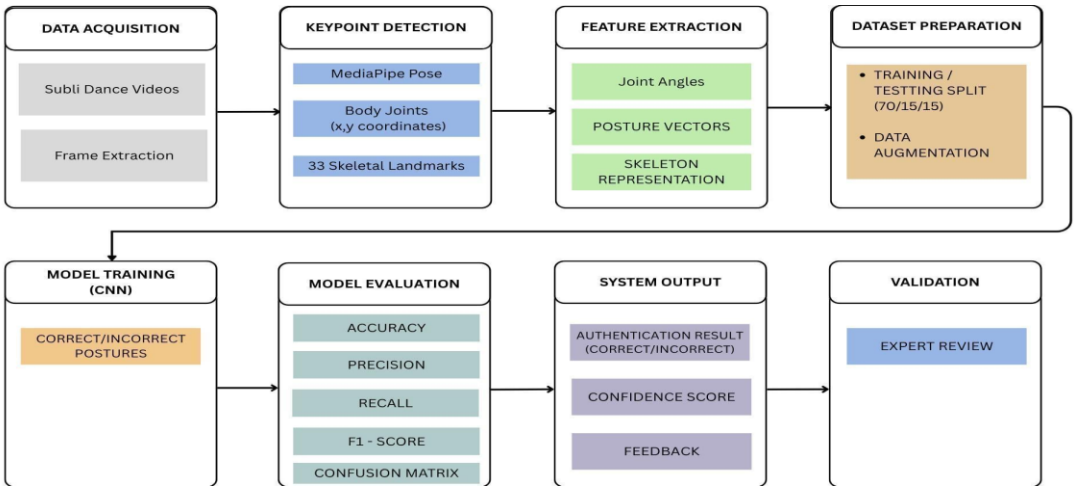


Figure 3. Model Development

Data Acquisition

The study gathered data through recorded video footage of Sublí dance performances conducted at Brgy. Bayog, Los Baños, Laguna. The primary subject was a local Sublí dancer who demonstrated authentic movements representative of traditional Filipino folk culture. To expand the dataset and ensure variation in posture and gesture, the researchers also participated in performing the Sublí dance themselves. Each recording session was carried out in a controlled setting with proper lighting, consistent framing, and stable camera positioning to maintain clarity and uniformity in the captured visuals. These procedures ensured that the dataset reflected accurate and analyzable representations of Sublí movements, including hand gestures, footwork, and overall body coordination.

The collected videos were converted into image frames and organized into separate subsets for model development. The dataset was divided into

70% for training, 15% for validation, and 15% for testing to optimize CNN's performance. The training set, comprising the majority of the data, was used to teach the model to recognize correct and incorrect Sublí postures based on keypoint patterns. The validation set was utilized to fine-tune model parameters and prevent overfitting, while the testing set evaluated the model's ability to generalize unseen data. This structured data partitioning ensured balanced learning and reliable performance assessment during the authentication of Sublí dance movements.

Data Sources

The dataset utilized in this study is composed of high-resolution images meticulously extracted from a series of recorded Subli dance performances, curated to capture the intricate nuances of this traditional art form. To ensure the development of a high-fidelity machine learning model, each extracted frame underwent a rigorous annotation process, categorizing movements into two distinct classes "correct" and "not correct". The "correct" label identifies authentic movements executed by professionally trained dances, characterized by precise postural alignments and traditional gestures, while the "not correct" label encompasses deviations, such as improper limb positioning or non-traditional body mechanics. To bolster the model's robustness and prevent overfitting, the source videos were filmed under diverse environmental conditions, including varying lighting intensities and multiple camera perspectives. This variety ensures that the Convolutional Neural Network (CNN) learns to prioritize movement patterns over background noise or environmental artifacts. Ultimately, this structured dataset serves as the

essential framework for training, validating, and testing the CNN, providing the model with the necessary data to accurately distinguish between authentic Subli traditions and erroneous executions.

Table 3. Dataset Description

Attribute	Description
Data Type	Images (JPEG) extracted from video
Source	Video Recording
Number of Images	419
Image Resolution	1080 x 1920 pixels
Video Length	1:42 minute
Video Resolution	2160 x 3840 pixels
Annotation	Keypoint-base (Mediapipe pose – 33 keypoints)
Environmental Condition	Indoor, Controlled lightning
Number of Dancer(s)	Expert: 1 Not Expert: 2
Number of Samples per Class	Correct: 214 Not Correct: 205
Number of Dance Categories	1

Model Selection

The Convolutional Neural Network (CNN) is the primary deep learning model utilized in this study for keypoint-based Sublí dance movement authentication. CNNs are perfect for visual posture analysis due to their ability to learn spatial feature hierarchies through multiple layers. In this study the CNN is used to classify dance poses based on extracted skeletal keypoints which enables the system to recognize and evaluate specific Sublí postures. Convolutional layers process joint spatial relationships pooling layers simplify

data while preserving important movement characteristics and fully connected layers perform final classification.

The dataset includes pose-estimated photos of the Sublí dance taken from a local dancer's video recordings in Brgy. Bayog Laguna and Los Baños in addition to the researchers own performances. The CNN is trained using these pose data to differentiate between legitimate Sublí poses and improper or irrelevant body positions. By digitally analyzing and validating traditional dance movements this model functions as the study's primary mechanism for posture verification supporting the larger objective of conserving Filipino cultural heritage.

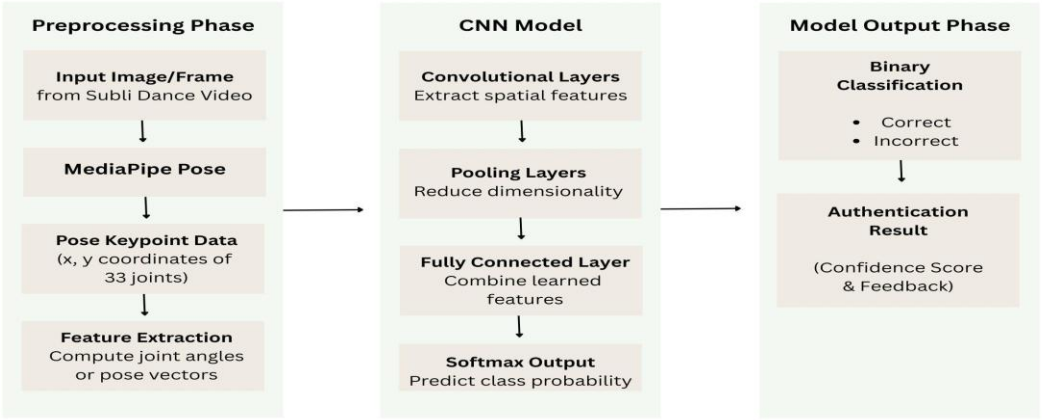


Figure 4. Model Architecture

Model Evaluation

To assess the performance of the developed CNN model for posture classification in Sublí dance, standard evaluation metrics for human posture-based classification will be applied. These include:

Accuracy. The overall percentage of correctly classified postures (e.g., “Correct” or “Incorrect”) out of all predictions made by the model.

Precision. For each class (particularly "Correct"), this measures the proportion of correctly predicted instances out of all instances predicted to belong to that class.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (1)$$

A high precision value indicates that the model makes few false positive errors.

Recall. Indicates how many actual instances of a class (e.g., correct postures) were correctly identified by the model.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (2)$$

A high recall value means the model is effective at detecting authentic dance movements.

F1-Score. The harmonic mean of precision and recall, providing a balanced assessment of model performance, particularly useful when dealing with class imbalance.

$$F1\ Score = 2 \cdot \frac{Pr\ ecision \cdot Recall}{Pr\ ecision + Recall} \quad (3)$$

Confusion Matrix. A visual representation of the classification results, showing the counts of true positives, true negatives, false positives, and false negatives. This allows for a more granular understanding of where the model succeeds or fails.

Loss Curves. Graphs showing the training and validation loss over epochs illustrate how the Sublí dance posture classification model learns during

training. These plots are used to monitor the model's learning behavior, identify signs of overfitting or underfitting, and ensure that the CNN achieves optimal generalization when authenticating unseen Sublí dance movements.

Model Validation

The generalization performance of the CNN-based posture analysis model for Sublí dance authentication was assessed through model validation. A distinct validation dataset from the training dataset was used to test the models correctness in identifying and classifying Sublí dance postures that it had never seen before. This process ensured that the model's functionality persisted after training and could be relied upon in real-world applications such as camera-based real-time posture recognition.

One of the validation steps was monitoring key performance metrics that the training framework automatically recorded. These metrics which are typical in CNN-based image classification assess the models capacity to learn and distinguish between authentic Sublí dance postures and other non-Sublí movements.

Validation Accuracy. Validation accuracy is defined as the proportion of dance postures in the validation dataset that are correctly classified. It illustrates how well CNN models generalize to fresh data. The model effectively learned the unique structural and visual characteristics of authentic Sublí movements as evidenced by a consistently high validation accuracy.

Validation Loss. Validation loss measures the prediction error of the model on unseen data. It is an essential indicator of how well the model

performs outside of the training set. Models with lower validation loss values are more capable of generalizing without overfitting. By tracking validation loss during training the model is protected from memorization of training samples and maintains balanced learning.

Training and Validation Curves. To illustrate the model's learning process, accuracy and loss curves for training and validation were plotted. These graphs helped assess overfitting and convergence, minimal divergence and steady improvement indicate consistent model performance.

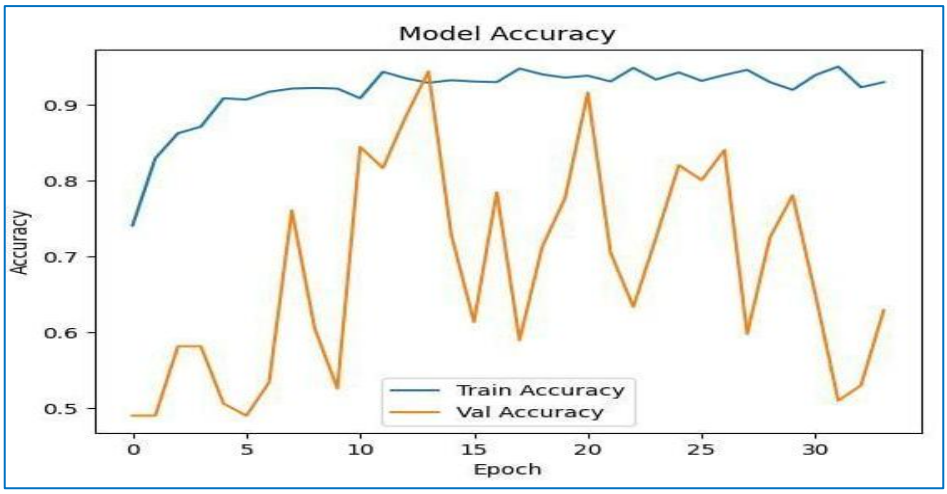


Figure 5. Training and Validation Accuracy

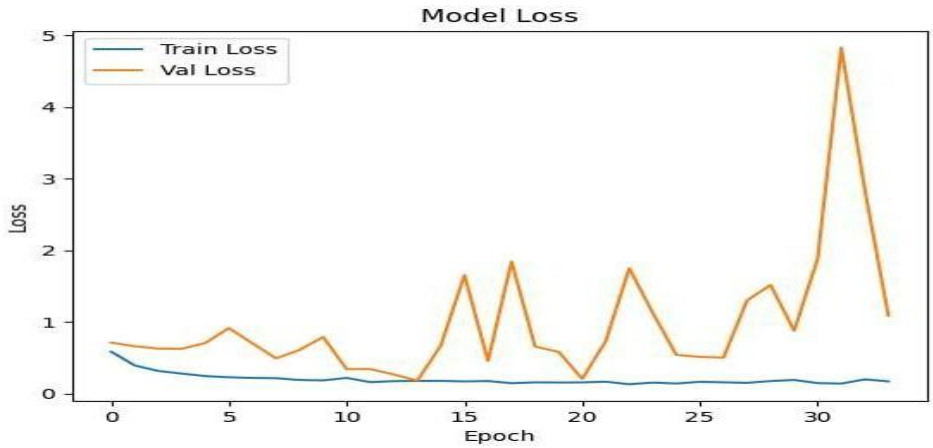


Figure 6. Training and Validation Loss

Confusion Matrix Analysis. By creating a confusion matrix, the model's classification performance was thoroughly examined. It shows the percentage of samples for each posture category that were correctly and incorrectly classified. The confusion matrix helps identify which specific Sublí postures were the hardest for the model to classify and which were the most correctly identified.

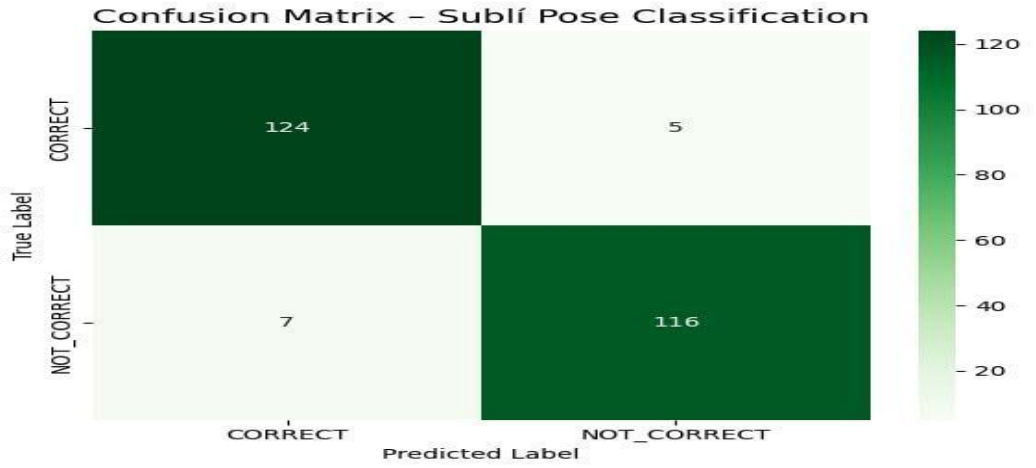


Figure 7. Confusion Matrix

T-Test Validation: A paired t-test was employed to statistically validate the performance of the developed Sublí Dance Movement Authentication Model. This statistical method was selected because both the model and the expert dancer evaluated the same set of dance posture samples, making their scores naturally paired. The test aimed to determine whether there was a significant difference between the model's classification outputs and the expert's evaluation of correct and incorrect Sublí dance movements. By comparing two related datasets, the CNN model's predictions and the expert's ground-truth assessment, the paired t-test served as the most appropriate analytical approach. Through this comparison, the analysis examined whether the differences in accuracy, confidence scores, or classification judgments for each specific sample were statistically meaningful rather than occurring by random variation. Conducting the paired t-test provided quantitative evidence of the

model's reliability, strengthening the claim that its performance is aligned with expert human judgment. Overall, the results supported the model's consistency and suitability for real-world application in authenticating Sublí dance postures.

$$t = \frac{d}{s_d/\sqrt{n}} \tag{4}$$

Ethical Considerations

This research will be conducted with strict adherence to ethical standards to ensure responsible research practices.

This research was conducted with strict adherence to ethical standards to ensure responsible research practices. The study complied with ethical guidelines for research involving human subjects and cultural expressions. Prior to data acquisition, informed consent was obtained from all participants, including the local Sublí dancer from Brgy. Bayog, Los Baños, Laguna. The nature, purpose, and scope of the study were clearly explained to the participants, ensuring that they understood how their dance performances and likeness would be used in the development and evaluation of the proposed system. Participation was entirely voluntary, and participants were informed of their right to withdraw at any stage without consequence.

To minimize dependency on community performers and to uphold respect for the cultural significance of the Sublí, the researchers also participated as subjects in the dataset. The dance was approached with sensitivity to its traditional, cultural, and spiritual roots, ensuring that movements were documented and analyzed in accordance with authentic practices and were not taken out of context or used for exploitative purposes.

All collected data, including recorded videos and pose-estimated images, were handled with confidentiality. No personally identifiable information was disclosed, and the data was used solely for academic research and cultural preservation purposes. The study actively avoided cultural misappropriation by engaging respectfully with the community, acknowledging cultural ownership, and ensuring that cultural knowledge was neither misused nor commercialized. Ethical clearance was obtained from the appropriate institutional body to confirm the study's compliance with established ethical research standards.

Preserving traditional folk dances such as the Sublí is a cultural and technological challenge in a modernized era. Expanding on the goals of this study, this research offers an AI-driven method that bridges the gap between tradition and innovation by authenticating Sublí dance movements using CNN-based posture analysis and keypoint-based image processing.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter outlines the study's objectives, results, and discussion, accompanied by visual representations of the findings. The study aims to develop an authenticating model for subli dance movements and posture analysis.

Cultural Data Acquisition and Preprocessing

A key component of this study was the data collection process, which served as the foundation for developing and evaluating the *Sublí* dance posture authentication model. The dataset was derived from authentic *Sublí* dance performances that demonstrated traditional and genuine movement patterns representative of the dance form. To expand the dataset and ensure sufficient pose diversity, the researchers also performed the *Sublí* dance themselves, carefully adhering to the traditional choreography and established movement sequences.

The performances were recorded at the UPLB Carillon Plaza using an iPhone 13 equipped with a high-definition camera capable of capturing videos at 1080p resolution (1920 × 1080 pixels). The device was mounted at an approximate height of 150 centimeters and positioned at a frontal angle relative to the performer. This configuration was determined to be optimal for capturing the dancer's full body and minimizing occlusions, ensuring that all major joints were visible for accurate pose detection and keypoint extraction.

Following data collection, manual verification and labeling of the extracted frames were conducted with reference to official *Sublí* instructional materials and performance videos. Each frame was reviewed to confirm that the dancer's entire body was visible and that the posture represented a distinct movement. To further ensure the accuracy and authenticity of the dataset, the labeled frames were validated by a *Sublí* dance expert, who confirmed that the identified postures accurately reflected traditional dance standards. This expert validation was crucial in maintaining the cultural integrity of the dataset and enhancing the reliability of the CNN model's posture classification results.

Table 4. Dataset Summary for Posture Classification

Label Category	Number of Frames	Description
Correct Posture	214	Frames showing accurate Subli dance form and alignment.
Incorrect Posture	205	Frames showing improper posture, misalignment, or incomplete movements.
Total	**419**	-------

Table 4 provides a detailed summary of the dataset utilized for the Subli dance posture classification task. The dataset is divided into two primary label categories: Correct Posture and Incorrect Posture. The Correct Posture category includes 214 frames that capture dancers performing accurate Subli dance forms, demonstrating proper alignment, balance, and execution of movements. In contrast, the Incorrect Posture category consists of 205 frames

depicting improper or misaligned body positions, as well as incomplete or inaccurate movement patterns.

In total, the dataset comprises 419 frames, representing a balanced distribution between correct and incorrect samples. This balance ensures that the classification model can effectively learn to distinguish between proper and improper dance postures without bias toward either class. The annotated frames serve as the foundational data for training and validating the proposed machine learning model aimed at automating the evaluation of *Subli* dance posture accuracy.



(a) (b)

Figure 8. Sample data of (a)Correct and (b)Not Correct

Figure 8(a) illustrates sample images categorized under the Correct Posture class, as summarized in Table 4. These frames represent instances where the dancer accurately performs the Subli dance movements with proper form, alignment, and balance. The images demonstrate consistent body positioning and adherence to the traditional dance posture standards. These correctly aligned frames, totaling 214 as shown in Table 4, serve as positive examples in the dataset and play a crucial role in training the posture classification model to recognize accurate dance execution. The visual samples

in this subset highlight the key physical cues and body orientations that define proper Subli dance form.

The consistent joint alignment and clear separation of upper-body and lower-body keypoints in these samples allow the CNN model to learn stable posture patterns, contributing to higher classification confidence when identifying correct Sublí dance movements.

Table 5. Summary of Preprocessing Parameters

Process	Description	Output Format
Frame Resizing	Images were automatically scaled and normalized by MediaPipe Pose to ensure uniform size.	Normalized x-y values (0-1 scale)
Color Conversion	Images were converted from BGR to RGB for accurate pose detection.	Consistent RGB color space
Keypoint Extraction	Extracted 33 landmarks (x, y) representing major body joints used as CNN features.	66 keypoints features + labels
Dataset Split	Divided into 70% training, 15% validation, and 15% testing.	1, 173 train / 251 validation / 252 test samples

Table 5 outlines the preprocessing parameters applied to the dataset prior to model training and evaluation. The preprocessing stage ensures data consistency, normalization, and compatibility with the neural network model.

The Frame Resizing process automatically scales and normalizes each image using MediaPipe Pose, ensuring a uniform frame size across all samples. The normalized x–y coordinate values (ranging from 0 to 1) enable standardized spatial representation of body landmarks for the model.

Color Conversion is applied to transform images from the BGR color format (used by OpenCV) to RGB, ensuring color consistency and accurate pose detection. This conversion standardizes the input color space, improving the robustness of subsequent feature extraction.

The Keypoint Extraction stage employs MediaPipe to detect 33 body landmarks, each represented by (x, y) coordinates corresponding to major body joints such as shoulders, elbows, hips, and knees. These keypoints, totaling 66 features per frame, are used as inputs for the convolutional neural network (CNN) along with their respective posture labels.

Finally, the Dataset Split divides the processed dataset into 70% training, 15% validation, and 15% testing subsets, corresponding to 1,173 training, 251 validation, and 252 testing samples. This partitioning ensures effective model training, fine-tuning, and performance evaluation on independent data subsets, reducing overfitting and enhancing generalization capability.

These preprocessing steps collectively enhance model performance by reducing input variability and emphasizing posture-related features rather than visual appearance. By standardizing keypoint coordinates and isolating skeletal information, the CNN can focus on learning joint relationships and posture alignment, which are critical for accurately distinguishing correct and incorrect Sublí dance movements.

Pose Detection and Keypoint Extraction

The pose detection component of the system employs MediaPipe Pose, a machine learning framework developed by Google, to identify and track 33

anatomical landmarks across the human body. These landmarks include critical joint positions such as the shoulders, elbows, hips, knees, and ankles, which collectively form the skeletal representation of the dancer's movement. Each detected landmark generates a set of x, y, and z coordinates, along with a visibility score that indicates the confidence level of the detection. This process effectively converts complex visual information from dance performances into a structured numerical format, known as a keypoint vector, which serves as the input for the Convolutional Neural Network (CNN). By transforming visual motion data into quantifiable features, the system enables precise analysis of body alignment and posture, forming the foundation for accurate Sublí dance authentication.



Figure 9. Detected Keypoint on a Subli Dancer

As shown in figure 9 the detected skeletal keypoints overlaid on a Sublí dancer using the MediaPipe Pose framework. The figure shows the localization of major body joints, including the head, shoulders, elbows, hips, knees, and

ankles, which collectively represent the dancer's posture during movement execution. The connected keypoints form a skeletal structure that captures body alignment, joint orientation, and limb positioning independent of background and clothing variations.

The accurate detection and alignment of these keypoints demonstrate the effectiveness of pose estimation in isolating posture-related features essential for Sublí dance authentication. By representing movements through skeletal landmarks, the system enables the CNN model to focus on joint relationships and posture correctness rather than visual appearance, thereby improving consistency and reliability in posture classification.

Hyperparameter Settings of CNN Model

The Convolutional Neural Network (CNN) model was trained using keypoint-based pose data derived from the MediaPipe Pose framework. These keypoints, which represent the spatial coordinates and visibility of major body joints, served as the primary input features for the network. The hyperparameters were carefully tuned to achieve an optimal balance between convergence speed, stability, and generalization performance. The model architecture adopted Custom CNN as its backbone due to its efficiency in extracting spatial and structural features from input data while maintaining lightweight computational requirements—making it suitable for real-time inference. The network was trained using the Adam optimizer, which adaptively adjusts learning rates for each parameter to accelerate convergence. The categorical cross-entropy loss function was employed to measure the discrepancy between predicted posture labels and ground truth data, enabling

the model to minimize classification errors effectively. This configuration provided a stable training process that supported both accuracy and generalization when classifying Sublí dance postures into correct or not correct categories.

Table 6. Hyperparameter Settings of the Posture Classification Model

Parameter	Description	Value
Optimizer	Optimization algorithm used during training.	Adam
Learning Rate	Step size for gradient descent updates.	0.001
Batch Size	Number of samples processed per training steps.	32
Epochs	Total iterations through the dataset.	50
Dropout Rate	Regularization to prevent overfitting.	0.3
Input Size	Number of keypoint features (33 x 2)	66
Model Backbone	CNN layers for classification (custom-built)	Custom CNN

Table 6 presents the selected hyperparameter settings used in training the CNN-based posture classification model. The Adam optimizer was employed due to its adaptive learning capability, enabling efficient convergence during training. A learning rate of 0.001 and a batch size of 32 were selected to balance training stability and computational efficiency. The model was trained for 50 epochs to allow sufficient learning without excessive overfitting, while a

dropout rate of 0.3 was applied as a regularization technique to reduce model dependency on specific neurons.

The input size of 66 features corresponds to the extracted 33 body keypoints represented by two-dimensional coordinates, ensuring that the model focuses solely on skeletal posture information. A custom CNN backbone was implemented to tailor the architecture specifically for Sublí dance posture classification.

Sample Implementation Overview

The developed system integrates pose detection, classification, and feedback generation within an interactive user interface built using the Gradio framework. This interface serves as the primary platform through which users can engage with the Sublí dance authentication model. It allows users to upload an image or video of a dancer performing Sublí movements, which the system then processes to analyze posture correctness. Upon submission, the system performs pose detection using MediaPipe Pose, extracts keypoint coordinates, and classifies the posture as either correct or not correct based on the trained CNN model. The results are then displayed in a user-friendly interface, providing both visual annotations and text-based feedback that highlight areas of improvement. This seamless integration of computer vision and machine learning within a web-based interface enables an accessible and practical approach to evaluating traditional dance movements, bridging the gap between cultural preservation and modern technology.

CNN-Based Posture Classification

The extracted keypoints are subsequently passed into the Convolutional Neural Network (CNN) for classification. Each keypoint vector, representing the spatial configuration of the dancer's body, is analyzed by the network to determine whether the posture corresponds to a Correct or Not Correct Sublí dance movement. The CNN processes these numerical representations to learn the spatial relationships and alignment patterns among the body joints, which are critical indicators of authentic dance execution. Through multiple convolutional layers, the model captures both local and global dependencies between landmarks, enabling it to distinguish subtle differences between proper and improper postures. This classification mechanism serves as the foundation of the system's authentication process, ensuring that each analyzed movement aligns with the traditional standards of Sublí choreography.

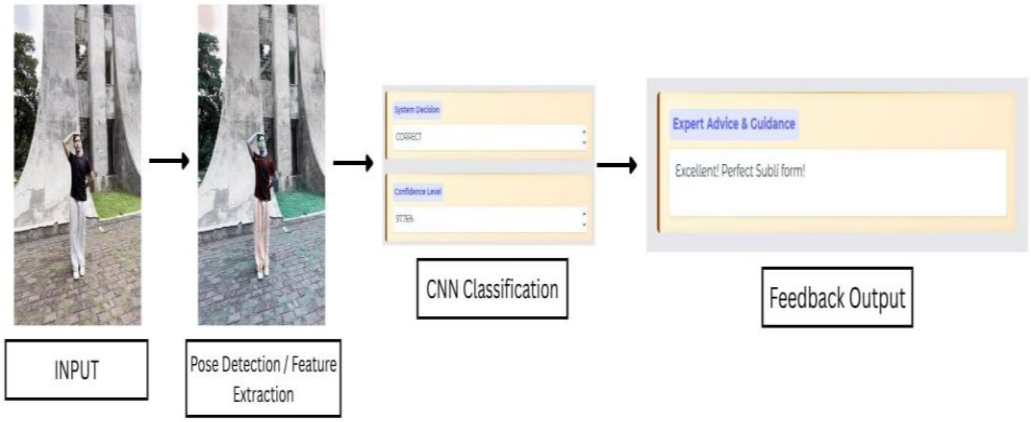


Figure 10. CNN Architecture for Subli Posture Classification

Figure 10 shows the overall architecture of the proposed CNN-based system for Subli dance posture classification and feedback generation. The process begins with an input image capturing a dancer performing a Subli posture. The image is then processed through a pose detection and feature

extraction stage, where key body joints are identified and converted into numerical features representing the dancer's skeletal structure.

These extracted features are forwarded to the CNN classification module, which learns spatial relationships among body keypoints to classify the posture as either CORRECT or NOT CORRECT. Based on the classification result, the system generates feedback output that provides either confirmation of correct posture execution or guidance on posture improvement.

This architecture was designed to prioritize skeletal posture information, which is essential in analyzing culturally significant dance movements such as Subli. The inclusion of pose detection ensures that the model focuses on joint alignment and body symmetry rather than irrelevant visual details like background or clothing. Using CNN allows the system to learn complex spatial dependencies between keypoints, such as arm positioning and torso alignment, which are critical indicators of correct Subli posture.

The integration of feedback output extends the system beyond simple classification by supporting instructional and corrective functions. This design choice aligns with the study's objective of not only authenticating Subli dance movements but also contributing to dancer training and cultural preservation. Overall, architecture supports accurate posture classification while maintaining interpretability and practical applicability.

Feedback and Scoring Mechanism

After the classification process, the system computes a pose score ranging from 0 to 100, which quantifies the overall quality of body alignment

based on the detected keypoints. This score reflects how closely the dancer's posture conforms to the correct Sublí form, serving as an interpretable measure of performance accuracy. Alongside the score, the system generates text-based feedback that focuses on specific anatomical regions such as the shoulders, arms, legs, and overall body balance. The feedback generation mechanism is implemented through a rule-based logic system, which evaluates spatial relationships among pose landmarks. For example, it assesses whether the shoulders are level, the arms are properly extended, and the stance width maintains balance. By combining quantitative scoring with qualitative feedback, the system not only identifies classification outcomes but also guides users toward posture correction and movement refinement.

Model Performance Evaluation

The system was evaluated using a set of unseen Sublí dance frames to assess its ability to accurately classify postures that were not part of the training dataset. Each image underwent a full analytical process, beginning with pose detection and keypoint extraction, followed by classification through the CNN model. The system determined whether the posture was Correct or Not Correct and computed a corresponding pose score to quantify alignment quality. Additionally, the framework generated text-based feedback for each test image, detailing specific movement corrections or affirmations based on body part positioning. This evaluation process allowed the researchers to observe the model's generalization capability and its effectiveness in providing actionable feedback for improving Sublí dance performance.

Table 7. System-Level Performance of the Posture Classification Model

Metrics	Precision	Recall	F1-Score	Support
Correct Posture	0.95	0.96	0.95	129
Not Correct Posture	0.96	0.94	0.95	123
Accuracy			**0.95**	**252**

Table 7 presents the system-level performance metrics of the posture classification model during testing. The model achieved an overall accuracy of 0.95%, indicating that it was able to correctly classify the majority of Sublí dance postures as either Correct or Not Correct. The precision and recall values for the Correct Posture class were 0.95 and 0.96, respectively, which demonstrates that the system effectively identifies correct poses while maintaining a relatively low rate of false positives. For the Not Correct Posture class, the model obtained a precision of 0.96 and a recall of 0.94, signifying that while the system can detect most misaligned postures, certain variations may still pose classification challenges. The F1-score, which represents the harmonic mean of precision and recall, recorded an average of 0.95, reflecting a balanced performance between sensitivity and specificity. Overall, these results confirm that the trained CNN model performs reliably when applied to unseen Sublí dance frames, effectively distinguishing between authentic and not correct postures for real-world applications.

The high accuracy and balanced precision–recall values indicate that the model successfully learned discriminative skeletal posture patterns relevant to Sublí dance movements. The strong recall for the Correct Posture class

suggests that expert-like joint alignments and arm positioning are consistently captured by the keypoint-based feature representation, enabling the model to recognize authentic postures with high confidence.

Meanwhile, the slightly lower recall observed in the Not Correct Posture class can be attributed to subtle posture deviations that closely resemble correct movements, particularly in static single-frame analysis. Since Sublí dance emphasizes smooth transitions and controlled motion, certain incorrect postures may still share similar spatial keypoint distributions with correct poses, leading to occasional misclassification.

The consistently high F1-score demonstrates that the model maintains a good balance between detecting correct postures and identifying incorrect ones. This confirms that the combination of pose-based feature extraction and CNN-based classification is effective for posture authentication tasks, supporting the system's suitability for instructional use, performance evaluation, and cultural preservation of Sublí dance.

Table 8. Sample Image Analysis Result

Step	Classification	Confidence Score	Feedback Summary
Step 1	Correct	100.00%	Excellent! Perfect Subli Form!
Step 2	Correct	90.14%	Excellent! Perfect Subli form!
Step 3	Correct	97.81%	Excellent! Perfect Subli form!
Step 4	Correct	97.81%	Excellent! Perfect Subli form!
Step 5	Correct	95.38%	Excellent! Perfect Subli Form!
Step 6	Correct	100.00%	Excellent! Perfect Subli Form!

Step 7	Correct	100.00%	Excellent! Perfect Subli Form!
Step 8	Correct	97.81%	Excellent! Perfect Subli form!
Step 9	Not Correct	68.81%	Needs improvement. Ensure your torso is properly positioned.
Step 10	Not Correct	81.66%	Needs improvement. Work on overall flow and balance of the pose.
Step 11	Not Correct	72.33%	Needs improvement. Ensure your torso is properly positioned.
Step 12	Correct	95.38%	Excellent! Perfect Subli Form!
Step 13	Correct	98.16%	Excellent! Perfect Subli Form!
Step 14	Correct	97.45%	Excellent! Perfect Subli Form!
Step 15	Correct	97.47%	Excellent! Perfect Subli Form!
Step 16	Correct	97.29%	Excellent! Perfect Subli Form!
Step 17	Correct	93.58%	c
Step 18	Correct	83.89%	Great! Just minor adjustment needed for perfection.
Step 19	Correct	96.36%	Excellent! Perfect Subli Form!
Step 20	Correct	66.71%	Basic form is correct but needs improvement.
Step 21	Correct	83.31%	Great! Just minor adjustments needed for perfection.
Step 22	Not Correct	60.00%	Needs improvement. Check your leg alignment and foot placement.
Step 23	Correct	83.74%	Great! Just minor adjustments needed for perfection.
Step 24	Correct	64.82%	Basic form Is correct but needs improvement.
Step 25	Correct	71.75%	Great! Just minor adjustments needed for perfection.
Step 26	Correct	99.45%	Excellent! Perfect Subli Form!

Step 27	Correct	92.62%	Excellent! Perfect Subli Form!
Step 28	Correct	91.85%	Excellent! Perfect Subli Form!
Step 29	Correct	83.10%	Great! Just minor adjustments needed for perfection.
Step 30	Correct	72.45%	Great! Just minor adjustments needed for perfection.
Step 31	Correct	95.92%	Excellent! Perfect Subli form!
Step 32	Correct	83.89%	Great! Just minor adjustment needed for perfection.
Step 33	Correct	97.69%	Excellent! Perfect Subli form!
Step 34	Correct	100.00%	Excellent! Perfect Subli form!
Step 35	Correct	99.68%	Excellent! Perfect Subli form!
Step 36	Correct	99.22%	Excellent! Perfect Subli form!
Step 37	Correct	86.50%	Great! Just minor adjustments needed for perfection.
Step 38	Correct	74.30%	Great! Just minor adjustments needed for perfection.
Step 39	Correct	73.74%	Great! Just minor adjustments needed for perfection.
Step 40	Correct	97.88%	Excellent! Perfect Subli form!
Step 41	Correct	99.67%	Excellent! Perfect Subli form!
Step 42	Correct	99.62%	Excellent! Perfect Subli form!
Step 43	Correct	99.78%	Excellent! Perfect Subli form!
Step 44	Not Correct	60.00%	Needs improvement. Check your leg alignment and foot placement.
Step 45	Correct	98.96%	Excellent! Perfect Subli form!
Step 46	Correct	93.26%	Excellent! Perfect Subli form!
Step 47	Correct	77.76%	Great! Just minor adjustments needed for perfection.
Step 48	Correct	97.81%	Excellent! Perfect Subli form!
Step 49	Correct	99.86%	Excellent! Perfect Subli form!
Step 50	Correct	99.75%	Excellent! Perfect Subli form!

Step 51	Correct	88.84%	Great! Just minor adjustments needed for perfection.
Step 52	Correct	99.51%	Excellent! Perfect Subli form!
Step 53	Correct	99.17%	Excellent! Perfect Subli form!
Step 54	Correct	95.99%	Excellent! Perfect Subli form!
Step 55	Correct	72.34%	Great! Just minor adjustments needed for perfection.
Step 56	Correct	93.50%	Excellent! Perfect Subli form!
Step 57	Correct	99.33%	Excellent! Perfect Subli form!
Step 58	Correct	94.68%	Excellent! Perfect Subli form!
Step 59	Correct	66.41%	Basic form is correct but needs improvement.
Step 60	Correct	96.08%	Excellent! Perfect Subli form!
Step 61	Correct	95.92%	Excellent! Perfect Subli form!
Step 62	Correct	99.17%	Excellent! Perfect Subli form!
Step 63	Correct	97.21%	Excellent! Perfect Subli form!
Step 64	Correct	63.79%	Basic form is correct but needs improvement.
Step 65	Correct	97.26%	Excellent! Perfect Subli form!
Step 66	Correct	98.24%	Excellent! Perfect Subli form!
Step 67	Correct	85.20%	Great! Just minor adjustments needed for perfection.
Step 68	Correct	55.97%	Basic form is correct but needs improvement.
Step 69	Correct	98.10%	Excellent! Perfect Subli form!
Step 70	Correct	73.07%	Great! Just minor adjustments needed for perfection.
Step 71	Correct	84.42%	Great! Just minor adjustments needed for perfection.
Step 72	Correct	96.08%	Excellent! Perfect Subli form!
Step 73	Correct	94.90%	Excellent! Perfect Subli form!
Step 74	Correct	97.66%	Excellent! Perfect Subli form!
Step 75	Correct	98.38%	Excellent! Perfect Subli form!
Step 76	Correct	98.19%	Excellent! Perfect Subli form!
Step 77	Correct	99.36%	Excellent! Perfect Subli form!

Step 78	Correct	97.49%	Excellent! Perfect Subli form!
Step 79	Correct	82.08%	Great! Just minor adjustments needed for perfection.
Step 80	Correct	86.89%	Great! Just minor adjustments needed for perfection.
Step 81	Correct	92.93%	Excellent! Perfect Subli form!
Step 82	Correct	89.69%	Great! Just minor adjustments needed for perfection.
Step 83	Correct	97.58%	Excellent! Perfect Subli form!
Step 84	Correct	91.23%	Excellent! Perfect Subli form!
Step 85	Correct	95.53%	Excellent! Perfect Subli form!
Step 86	Correct	98.92%	Excellent! Perfect Subli form!
Step 87	Correct	88.14%	Great! Just minor adjustments needed for perfection.
Step 88	Correct	89.37%	Great! Just minor adjustments needed for perfection.
Step 89	Correct	82.32%	Great! Just minor adjustments needed for perfection.
Step 90	Correct	88.68%	Great! Just minor adjustments needed for perfection.
Step 91	Correct	87.53%	Great! Just minor adjustments needed for perfection.
Step 92	Correct	74.86%	Great! Just minor adjustments needed for perfection.
Step 93	Correct	95.26%	Excellent! Perfect Subli form!
Step 94	Correct	78.01%	Great! Just minor adjustments needed for perfection.
Step 95	Correct	93.01%	Excellent! Perfect Subli form!
Step 96	Correct	89.37%	Great! Just minor adjustments needed for perfection.
Step 97	Correct	100.00%	Excellent! Perfect Subli form!
Step 98	Correct	86.25%	Great! Just minor adjustments needed for perfection.

Step 99	Correct	57.22%	Basic form is correct but needs improvement.
Step 100	Correct	73.50%	Great! Just minor adjustments needed for perfection.
Step 101	Correct	85.30%	Great! Just minor adjustments needed for perfection.
Step 102	Correct	87.34%	Great! Just minor adjustments needed for perfection.
Step 103	Correct	90.16%	Excellent! Perfect Subli form!
Step 104	Correct	100.00%	Excellent! Perfect Subli form!
Step 105	Correct	57.22%	Basic form is correct but needs improvement.
Step 106	Correct	90.14%	Excellent! Perfect Subli form!

Table 8 presents a set of sample results generated from the system's image analysis testing, showing both the Correct and Not Correct executions for each Sublí dance step. For every step, the model processed paired posture samples, one performed correctly and one performed incorrectly, to evaluate how accurately the CNN-based system distinguishes authentic movement execution from posture deviations. Each processed image includes the model's classification output, confidence score, pose score, and feedback summary.

The classification output identifies whether the detected posture is recognized as Correct or Not Correct based on the model's analysis of keypoint alignment. The confidence score reflects the level of certainty in the system's decision, while the pose score (0–100) quantifies the overall form quality and posture alignment for each sample. As shown in the table, correctly executed postures consistently achieved higher confidence and pose scores, demonstrating proper alignment, stable body positioning, and adherence to Sublí movement characteristics. In contrast, their corresponding "Not Correct"

counterparts received lower scores, flagging noticeable deviations in key areas such as shoulder height, arm extension, torso balance, or stance width.

The feedback summary provides targeted, step-specific suggestions derived from the keypoint errors detected. For each step, the system was able to pinpoint what needed correction, whether it was elbow elevation, hip positioning, or weight distribution, making the output both diagnostic and actionable. By presenting Correct and Not Correct samples side-by-side for every step, Table 8 effectively illustrates the model's ability to (1) recognize proper Sublí posture execution, (2) detect and label deviations, and (3) generate meaningful corrective feedback. This demonstrates the system's reliability not only in posture classification but also in supporting dancer improvement through precise, interpretable guidance.

Statistical Analysis (Paired T-test)

Table 9. Results of the Paired T-test Between Correct and Not Correct Subli Dance Posture

T-Test	Statistic	df	p	Mean difference	SE difference	Effect Size
Paired T-Test	-0.445	102	0.657	-0.0971	0.0218	-0.0439

Note. H_a μ Measure 1 - Measure 2 ≠ 0

As shown in Table 9, a paired t-test was conducted to examine whether there was a statistically significant difference between the system's two measurement outputs, Measure 1 and Measure 2, which repres5ent the model's paired evaluations of Sublí dance postures. The analysis yielded a t-

statistic of –0.445 with 102 degrees of freedom and a p-value of 0.657, which is greater than the 0.05 significance level. This indicates that there is no significant difference between the two measurement outputs produced by the system.

The mean difference (–0.0971) shows that the score from Measure 1 was slightly lower than that of Measure 2, although the variation is minimal and statistically insignificant. The standard error of the difference (0.0218) also suggests that the paired outputs are consistent across samples. Additionally, the effect size (–0.0439) indicates a very small practical difference, reinforcing that the two measures are essentially equivalent.

These results imply that the system's evaluation of Sublí postures remains stable and internally consistent, regardless of which measurement instance is considered. Since the two outputs do not differ significantly, the model demonstrates reliable scoring behavior when analyzing the same movement inputs. This consistency is important for authentication, as it confirms that the keypoint-based feature extraction and CNN-based posture classification generate repeatable and unbiased results. The lack of significant difference between the two measures supports the conclusion that the system maintains uniform interpretation of posture quality, contributing to its overall credibility in authenticating Sublí dance movements.

Descriptive

Table 10. Descriptive Comparison of Correct and Not Correct Subli Dance Posture

	N	Mean	Median	SD	SE
A	103	0.961	1	0.194	0.0191
B	103	0.971	1	0.169	0.0167

Table 10 presents the descriptive statistics of the model's confidence scores for the two posture categories. Group A and Group B each consist of 103 samples, indicating a balanced dataset used for analysis. The mean confidence score for Group A is 0.961, while Group B exhibits a slightly higher mean of 0.971. Both groups share a median value of 1.0, suggesting that most predictions were made with high confidence by the CNN model.

In terms of variability, Group A has a standard deviation of 0.194, whereas Group B shows a lower standard deviation of 0.169, indicating more consistent confidence scores in Group B. The standard error values of 0.0191 for Group A and 0.0167 for Group B further suggest minimal sampling error and stable mean estimates. Overall, these descriptive results indicate that the model produced consistently high confidence scores for both posture categories, supporting the reliability of the CNN-based classification prior to inferential statistical testing.

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary of the Study

The Sublí dance, a traditional devotional performance from Batangas, reflects the cultural and spiritual heritage of the Filipino people. Its intricate movements and postures, passed down through generations, present a challenge for modern documentation and teaching, particularly in ensuring authenticity. This study addressed this challenge by developing a keypoint-based Convolutional Neural Network (CNN) model to analyze and classify Sublí dance movements as either Correct or Not Correct.

Authentic Sublí performances were recorded at the UPLB Carillon Plaza, and the videos were processed into individual image frames. Using the MediaPipe Pose framework, the system extracted skeletal keypoints representing the dancer's body posture. These keypoints were converted into structured numerical datasets, providing standardized inputs for the CNN-based classification model and ensuring consistency and precision across all samples.

The CNN model was trained to recognize spatial relationships among joint angles and body alignments characteristic of traditional Sublí movements. Evaluation using accuracy, precision, recall, and F1-score showed high performance and consistency, confirming the model's effectiveness in distinguishing Correct from Not Correct postures.

Additionally, a paired t-test was conducted to assess the consistency of the system's measurement outputs, yielding no significant difference between paired evaluations ($p = 0.657$). This demonstrates that the model produces stable, reliable, and unbiased results across repeated assessments.

Overall, the results demonstrate that the developed keypoint-based CNN system can accurately and reliably authenticate Sublí dance movements. By integrating image processing and deep learning techniques, this study provides a technological tool for preserving, analyzing, and teaching traditional Filipino dances, supporting the continued authenticity and transmission of Sublí movements in the modern digital era.

Conclusions

This section summarizes the study's conclusion. Each research objective is systematically addressed to provide a clear interpretation of the results and confirm alignment with research goals.

1. **To develop and train a CNN model for classifying Sublí dance movements as "Correct" or "Not Correct".**

The design, development, and training Convolutional Neural Network (CNN) model enabled the system to accurately classify Sublí dance movements as Correct or Not Correct. The CNN was structured to process pose-based features derived from the extracted keypoint data, allowing the model to learn complex spatial relationships between body joints. During training, the model was exposed to labeled posture samples representing both authentic and incorrect Sublí movements, enabling it to recognize distinguishing patterns in body alignment, limb positioning, and overall posture structure. The training process optimized the model's parameters using

backpropagation and an adaptive optimizer, improving classification accuracy over successive epochs. Through this approach, the CNN effectively learned to generalize posture patterns, forming a reliable classification model for authenticating Sublí dance movements.

2. **To implement the trained CNN model with keypoint detection on the extracted images.**

After implementing the trained Convolutional Neural Network (CNN) together with the keypoint detection process, the system was able to classify extracted Sublí dance images as Correct or Not Correct with high reliability. Using MediaPipe Pose, key body landmarks were detected from each image and converted into normalized coordinate data, which served as input to the trained CNN model. By analyzing the spatial relationships and geometric configurations of these keypoints, the model effectively distinguished between authentic and incorrect Sublí postures.

3. **To evaluate its performance using accuracy, precision, recall, and F1-Score.**

The performance of the implemented model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results demonstrated strong and consistent performance across both posture categories, achieving an overall accuracy of 0.95. High precision and recall values for both classes indicate that the model accurately identified correct postures while minimizing misclassification of incorrect movements. These findings confirm that the trained CNN model generalizes well to unseen

data and provides a reliable framework for authenticating Sublí dance movements.

4. **To conduct a statistical analysis using paired t-test to determine significant differences between correct and not correct Sublí movements.**

The statistical analysis using a paired t-test was conducted to determine whether there was a significant difference between the system's two measurement outputs for Correct and Not Correct Sublí movements. This analysis aimed to assess the internal consistency and reliability of the model's evaluations across paired observations. The computed t-value of –0.445 with 102 degrees of freedom yielded a p-value of 0.657, which exceeds the 0.05 level of significance. This indicates that there was no statistically significant difference between the paired measurement scores generated by the system.

The result implies that the model maintained stable and consistent outputs when assessing Sublí postures, demonstrating that repeated evaluations produced uniform results regardless of classification type. This stability confirms that the developed system performs reliably and does not introduce variation or bias between paired assessments, strengthening its validity as an effective tool for authenticating Sublí dance movements through keypoint-based image processing and CNN-based posture analysis.

Recommendations

The following recommendations are presented for future research and the practical application of key point-based posture analysis in the preservation and education of traditional Filipino dances.

1. Future studies should expand the dataset by including more Sublí performers, varied camera angles, and additional posture samples to improve the model's generalization across different performers and environments.
2. Future researchers are encouraged to use temporal or sequence-based models, such as LSTM or temporal CNNs, to better capture the sequential nature of dance movements and improve recognition accuracy.
3. The developed system may be integrated into a mobile or web-based platform to serve as an interactive educational tool. Such applications can assist in teaching Sublí dance to younger generations, promoting cultural awareness and engagement.
4. The system can be enhanced for real-time detection of Sublí movements, providing immediate feedback to help dancers correct postures and improve training efficiency.
5. Finally, the study's methodology can be applied to other indigenous and folk dances in the Philippines, expanding cultural preservation through AI and computer vision.

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APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactory completed all the requirements of the thesis project entitled:

“AUTHENTICATING SUBLI DANCE MOVEMENTS THROUGH KEYPOINT-BASED IMAGE PROCESSING AND CNN-BASED POSTURE ANALYSIS”

ROXANNE M. REYES and CRISELLE UMALI have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

SHERWIN B. SAPIN, EdD
Thesis Project Adviser
November 13, 2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The thesis entitled **"AUTHENTICATING SUBLI DANCE MOVEMENTS THROUGH KEYPOINT-BASED IMAGE PROCESSING AND CNN-BASED POSTURE ANALYSIS"** prepared and submitted by Roxanne M. Reyes and Criselle Umali in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent Systems is at this moment recommended for approval and acceptance.

SHERWIN B. SAPIN, EdD
Thesis Project Adviser

APPENDIX B

Letter of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

Laguna State Polytechnic University – Los Baños Campus
College of Computer Studies
Brgy. Malinta, Los Baños, Laguna
June 20, 2025

ACCEPTANCE LETTER

Dear Mr. Bautista,

We are pleased to inform you that you have been officially chosen as a participant in our research study entitled **"Authenticating Subli Dance Movements Through Keypoint-Based Image Processing and CNN-Based Posture Analysis."** This study is being conducted by Roxanne M. Reyes & Criselle Umali, Bachelor of Science in Computer Science students from Laguna State Polytechnic University – Los Baños Campus.

Your participation as one of the **Subli dancers** will greatly contribute to the success of this study, which aims to analyze and authenticate traditional Subli dance movements through modern image processing and machine learning techniques. The data collected from your performances will be used solely for academic and research purposes and will be treated with strict confidentiality. We appreciate your willingness to be part of this research. Your expertise and cooperation are invaluable in preserving and promoting Filipino cultural heritage through technology.

Thank you and congratulations on being part of our study.

Sincerely,

Roxanne M. Reyes
Researcher

Sherwin B. Sapin, EdD
Adviser

Criselle Umali
Researcher

Jhed Greg O. Bautista
Subli Dancer

APPENDIX C

Relevant Source Code

Convolutional Neural Networks (CNN)

Pose Extraction Module

```
# MediaPipe Pose Initialization
#This Script Extracts Keypoints from Subli dance videos using MediaPipe Pose

import cv2
import mediapipe as mp
import csv

mp_pose = mp.solutions.pose
pose = mp_pose.Pose

cap = cv2.VideoCapture('subli_video.mp4')
with open('pose_keypoint.csv', 'w', newline='') as if:
        writer = csv.writer(f)
        while cap.isOpened():
                ret, frame = cap.read()
        if not ret:
                break
        results = pose.process(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        if results.pose_landmarks:
                keypoints = []
                for lm in results.pose_landmarks.landmark:
                        keypoints.extend([lm.x, lm.y])
                writer.writerow(keypoints)
cap.release()
```

The script initializes the MediaPipe Pose solution and utilizes OpenCV to extract body keypoints from a Sublí dance video for posture analysis. It begins by importing the necessary libraries—OpenCV for handling video input, MediaPipe for pose estimation, and CSV for saving extracted data. The MediaPipe Pose model is then initialized to detect human body landmarks from each frame of the video. Using OpenCV, the script reads the video file subli_video.mp4 frame by frame. Each frame is converted from BGR to RGB color format, as required by MediaPipe, and then processed to identify key pose landmarks. When landmarks are successfully detected, the script extracts their x and y coordinates and stores them as a list representing a single frame's keypoints. These coordinates are written to a CSV file named pose_keypoint.csv, allowing for structured storage of the captured movement data. Once all frames have been processed, the video capture is released. Overall, the script automates the extraction of pose keypoints from dance videos, producing a dataset that can be used for further analysis or as input to a machine learning model for posture authentication.

CNN Model Training

```
# CNN Model Training Module
```

```
# This script trains and evaluates a CNN model for classifying Sublí dance poses.

from tensorflow.keras.models Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout
from tensorflow.keras.preprocessing.image import ImageDataGenerator

model = Sequential([
        Conv2D(32, (3,3), activation='relu', input_shape=(224,224,3)),
        MaxPooling2D(2,2),
        Conv2D(64, (3,3), activation='relu'),
        MaxPooling2D(2,2),
        Flatten(),
        Dense(128, activation='relu'),
        Dropout(0.5),
        Dense(2, activation='softmax')
])
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy']

train_datagen = ImageDataGenerator(rescale=1./255, validation_split=0.2)
train_gen = train_datagen.flow_from_directory('/content/dataset', target_size=(224,224), batch_size=32, subset='training')
val_gen = train_datagen.flow_from_directory('/content/dataset', target_size=(224,224), batch_size=32, subset='validation')

history = model.fit(train_gen, validation_data=val_gen, epochs=20)
model.save('sublipose_model.h5')
```

The CNN Model Training Module is designed to train and evaluate a convolutional neural network for classifying Sublí dance postures. The script begins by importing essential Keras components, including model and layer classes such as Conv2D, MaxPooling2D, Flatten, Dense, and Dropout, as well as the ImageDataGenerator utility for data preprocessing. The model is built using the Sequential API, which stacks layers in a linear manner. It starts with convolutional layers that extract spatial features from the input images, followed by max-pooling layers that reduce dimensionality while preserving important features. After flattening the extracted features, fully connected (Dense) layers are applied for learning higher-level patterns, with a dropout layer introduced to prevent overfitting by randomly deactivating neurons during training. The output layer uses a softmax activation function to classify inputs into two categories, representing correct and incorrect Sublí poses. The model is compiled using the Adam optimizer and categorical cross-entropy loss function, with accuracy as the primary evaluation metric. The dataset, located in the /content/dataset directory, is processed using ImageDataGenerator, which rescales image pixel values and automatically divides the data into training (80%) and validation (20%) subsets. The model is then trained for 20 epochs, during which it learns to recognize and differentiate between correct and incorrect postures based on

image features. Finally, the trained model is saved as sublipose_model.h5 for future use in classification or real-time evaluation tasks.

Authentication Interface (Gradio App)

```
# Authentication Interface Module
# This Gradio-based interface allows real-time prediction and feedback

import gradio as gr
from tensorflow.keras.models import load_model
import numpy as np
import cv2
import mediapipe as mp

model = load_model('sublipose_model.h5')
mp_pose = mp.solutions.pose

def predict_pose(image):
    pose = mp_pose.Pose()
    results = pose.process(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    if not results.pose_landmarks:
        return "No pose detected."
    keypoints = []
    for lm in results.pose_landmarks.landmark:
        keypoints.extend([lm.x, lm.y])
    input_array = np.array(keypoints).reshape(1, -1)
    prediction = model.predict(input_array)
    label = "Expert" if np.argmax(prediction) == 0 else "Not Expert"
    return f"Predicted Class: {label}"

iface = gr.Interface(fn=predict_pose, inputs="image", outputs="text",
                     title="Sublí Dance Authentication")
iface.launch()
```

This script implements an Authentication Interface Module that provides a real-time graphical interface for evaluating Sublí dance postures using a trained CNN model. It integrates Gradio for the user interface, MediaPipe for pose detection, and OpenCV for image processing. The system loads a pre-trained model to classify uploaded images as Expert or Not Expert Sublí poses. Input images are processed by extracting pose keypoint coordinates, which are then fed into the model for prediction. If no pose is detected, the system returns an appropriate message. The Gradio interface allows users to upload images and view classification results through an accessible web-based platform, enabling interactive Sublí dance authentication.

APPENDIX D

Evaluation Tool

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

AUTHENTICATING SUBLI DANCE MOVEMENTS THROUGH KEYPOINT-BASED IMAGE PROCESSING AND CNN-BASED POSTURE ANALYSIS

Roxanne M. Reyes, Criselle Umali
Researchers

The primary goal of this research is to develop a tool capable of authenticating Sublí dance movements using image processing and machine learning techniques. This is accomplished by integrating keypoint-based pose estimation and a Convolutional Neural Network (CNN) model to analyze and classify dance postures as correct or incorrect, thereby supporting the preservation and accurate recognition of traditional Sublí dance movements.

This questionnaire, referred to in our study as the Evaluation Tool, is designed to help us evaluate the model's performance in classifying Subli dance movements. Your honest evaluation and expert insights are essential for validating and refining our model.

Our system acts as a posture authentication tool that analyzes recorded Subli dance movements through body keypoints and posture patterns.

Here's how it works in three simple steps:

1. Pose Detection (MediaPipe)
    - The system processes the dance video and uses MediaPipe Pose to detect and extract body keypoinst corresponding to the dancer's major joints and landmarks.
    - Result: The system maps a skeletal keypoint overlay on the dancer's image, forming the basis for movement analysis.
2. Feature Extraction
    - The detected keypoints are converted into numerical data (x, y, z coordinates and visibility) which represent the position of each body part in every frame.
    - Result: This step transforms visual dance movements into measurable data for analysis.
3. Posture Classification (CNN Model)

- The extracted data are fed into a CNN Model trained to recognize whether a particular posture or movement is correct or incorrect based on expert demonstrations of the Subli dance.

The classification process involves several stages:

- **Convolutional Layers:** Capture the spatial relationships between body keypoints to identify posture-specific patterns.
- **Pooling Layers:** Reduce data complexity while preserving essential movement features.
- **Fully Connected Layers:** Integrate and analyze extracted features to form a holistic interpretation of the posture.
- **Output Layer:** Produces a final classification label (Correct or Incorrect) along with a confidence score (ranging from 0 to 1).
- Result: The CNN Model evaluates the detected pose and provides an objective classification and confidence level indicating movement correctness.

Your role as the expert validator is to assess the system's outputs and verify if the predicted classifications correctly represent authentic Subli dance movements based on traditional form and proper execution.

Validation Procedure

The purpose of this validation is to assess the model's performance in accurately classifying Subli dance movements.

The validation process will involve the following steps:

1. **Data Provision:** Our research team will provide you with a minimum of 100 images or video frames representing various Subli dance postures that were classified by the developed model.

2. **Expert Validation:** As the validator, your expertise will determine whether the classification shown (correct or incorrect) accurately reflects the authenticity of the movement according to traditional Sublí standards.

3. **Performance Marking:** Following your review, a scoring table will be provided for you to mark the model's performance on a per-clip/image basis:

- A score of 1 will be assigned if the model's classification (the highlighted keypoints and label) correctly identifies a Sublí dance movement as *correct* based on your expert validation.
- A score of 0 will be assigned if the model's classification (the highlighted keypoints and label) does not accurately represent the Sublí dance movement, meaning the model's result differs from your expert evaluation of whether the posture is correct or incorrect.

This validation will ultimately establish the degree of concordance between the model's prediction and your expert assessment.

Please be assured that all your responses will be kept strictly confidential and used solely for academic purposes.

VALIDATION TABLE

Instruction:

Performance Marking: Following your review, this scoring table is provided for you to mark the model's performance on a per-clip/image basis:

- A score of 1 will be assigned if the model's classification (the highlighted keypoints and label) correctly identifies a Sublí dance movement as *correct* based on your expert validation.

- A score of 0 will be assigned if the model's classification (the highlighted keypoints and label) does not accurately represent the Sublí dance movement, meaning the model's result differs from your expert evaluation of whether the posture is correct or incorrect.

Images	Score
Image 1	1
Image 2	1
Image 3	1

Images	Score
Image 108	1
Image 109	1
Image 110	1

Image 4	1
Image 5	1
Image 6	1
Image 7	1
Image 8	1
Image 9	0
Image 10	1
Image 11	1
Image 12	1
Image 13	1
Image 14	1
Image 15	1
Image 16	1
Image 17	1
Image 18	1
Image 19	1
Image 20	1
Image 21	0
Image 22	1
Image 23	1
Image 24	1
Image 25	1
Image 26	1
Image 27	1
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Image 111	1
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Image 121	1
Image 122	1
Image 123	1
Image 124	1
Image 125	1
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Image 37	1	Image 144	1
Image 38	1	Image 145	1
Image 39	1	Image 146	1
Image 40	1	Image 147	1
Image 41	1	Image 148	1
Image 42	1	Image 149	1
Image 43	1	Image 150	1
Image 44	1	Image 151	1
Image 45	1	Image 152	1
Image 46	1	Image 153	1
Image 47	1	Image 154	1
Image 48	1	Image 155	1
Image 49	1	Image 156	1
Image 50	1	Image 157	1
Image 51	1	Image 158	1
Image 52	1	Image 159	1
Image 53	1	Image 160	1
Image 54	1	Image 161	1
Image 55	1	Image 162	1
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Image 59	1	Image 166	1
Image 60	1	Image 167	1
Image 61	1	Image 168	0

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Image 204	1
Image 205	1

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Image 99	1	Image 206	1
Image 100	1	Image 207	1
Image 101	1	Image 208	1
Image 102	1	Image 209	1
Image 103	1	Image 210	1
Image 104	1	Image 211	1
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Image 107	1	Image 214	1



MARCO ANTONIO E. VALE, LPT
Signature over Printed Name
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APPENDIX E

Sample Input / Output

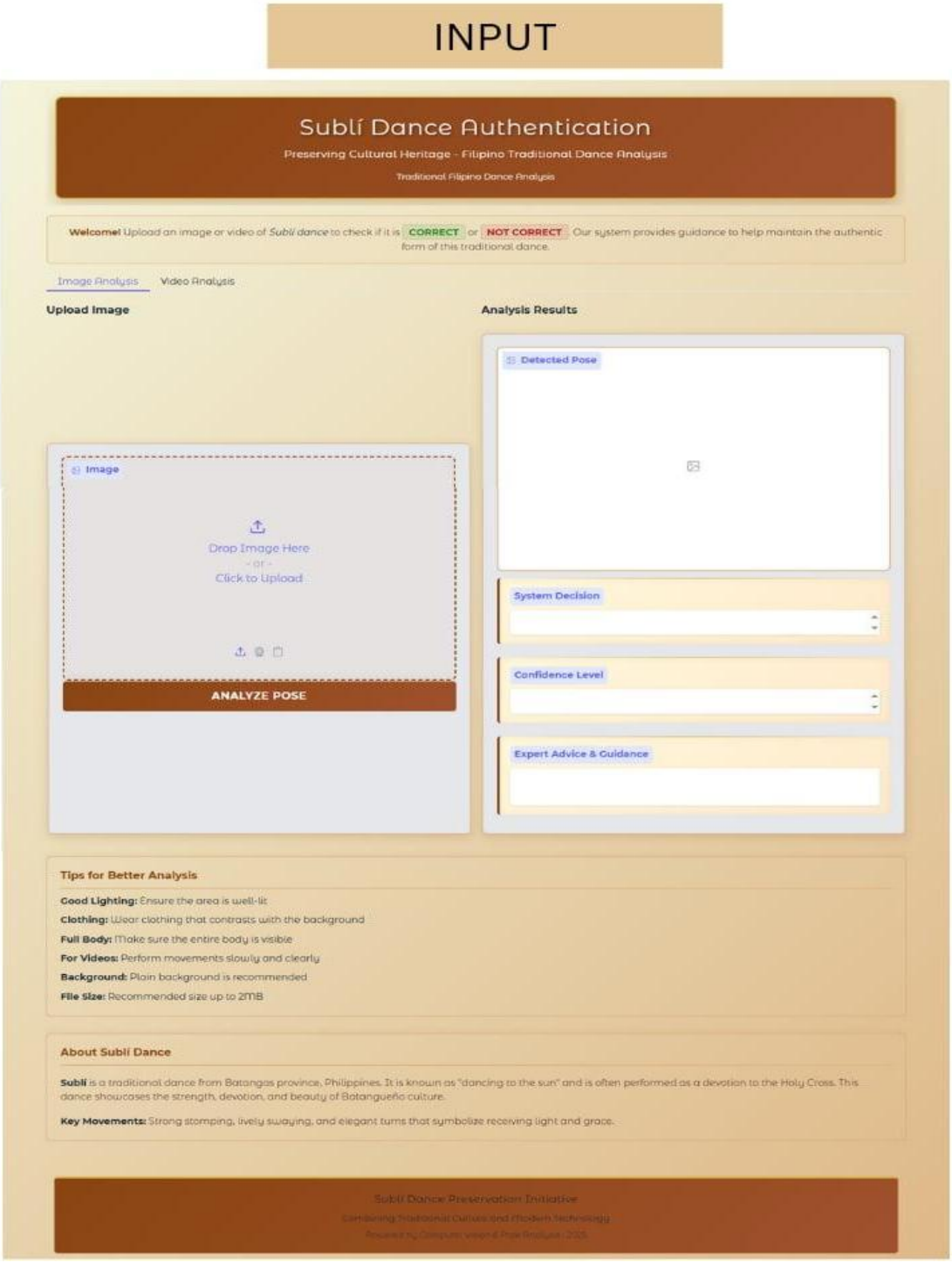


Figure 11. Input of the Subli Dance movements authentication

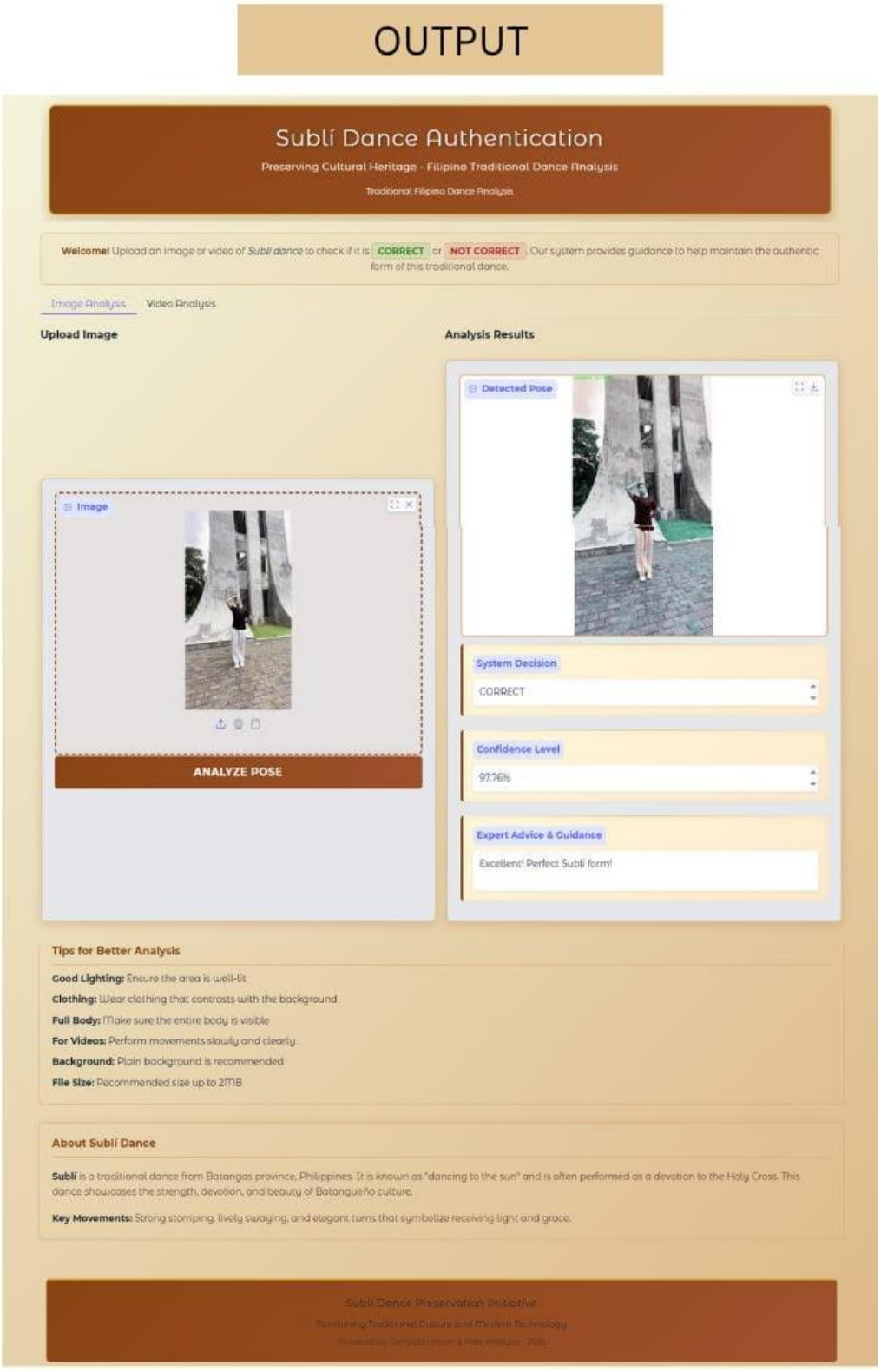


Figure 12. Output of the Subli Dance movements authentication

The figures illustrate the process flow of the Sublí Dance Authentication System, from the initial input stage to the final output generated by the model.

Figure 11 presents the Input Interface, where the user uploads an image or video frame of a Sublí dancer to be analyzed. The uploaded file is displayed in the “Upload Image” panel, and the analysis begins once the “Analyze Pose” button is clicked. This interface also provides Tips for Better Analysis, which include ensuring proper lighting, visible full-body posture, and a plain background to enhance detection accuracy. This step serves as the starting point of the system’s authentication process, where data is prepared and forwarded to the CNN model for analysis.

Figure 12 shows the Output Interface, which displays the results after the uploaded image has been processed by the CNN model. The “Detected Pose” section presents the analyzed image with overlaid keypoints representing the dancer’s body joints. The system then outputs the System Decision, identifying whether the detected posture is Correct or Not Correct, along with the Confidence Level, which quantifies the model’s certainty in its classification. Additionally, the Expert Advice & Guidance box provides feedback regarding the dancer’s performance, in this example, the system classified the movement as Correct with 97% confidence, accompanied by the remark “Excellent! Perfect Sublí form!”. Together, these figures demonstrate the full operational workflow of the system, from user input to analytical output, showcasing how keypoint-based image processing and CNN classification are integrated to authenticate Sublí dance postures accurately and efficiently.

APPENDIX F

User's Guide

HOW TO USE THE SYSTEM
SUBLI DANCE AUTHENTICATION SYSTEM

Preserving Cultural Heritage through Image Processing and CNN-Based Posture Analysis



1 LOADING PAGE

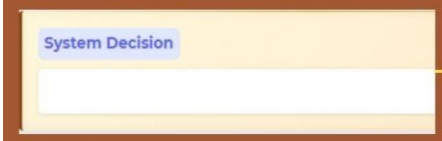
On the starting page, users prompted to upload an image or video of a Subli dance pose, Click ***Click to Upload or Simply drag and drop Image**



2 IMAGE ANALYSIS

After the image is uploaded, Click ***Analyze Pose.** The system will process the image using keypoint detection and CNN - Based analysis





3 SYSTEM DESICION

After the analysis, the system automatically classifies the dance movement as either:
CORRECT – matches the authentic Subli form
NOT CORRECT – deviates from the authentic movement
This classification is based on the trained CNN model that compares the uploaded pose to real Subli datasets.

4 CONFIDENCE LEVEL

The system displays a confidence percentage (e.g., 97.62%) that shows how certain the model is about its classification.





5 EXPERT ADVISE & GUIDANCE

After classification, the system provides automated feedback to help users improve their dance posture.
Examples of feedback include:
"Excellent! Perfect Subli form."
"Adjust arm position for better alignment."

REYES ROXANNE M. UMALI CRISELLE

APPENDIX G

Revision Matrix

MATRIX FOR REVISIONS

Commentors	Consultants	Comments	Page No.	Actions	Page No.
Mr. Gene Marck B. Catedrilla	Statistician	Abstract Suggested Improvement: Explicitly state final accuracy / performance metric in one sentence.	v	The abstract was revised and the final accuracy and performance metrics were stated in one sentence.	v
Mr. Gene Marck B. Catedrilla	Statistician	Keywords Consider adding: Human Pose Estimation, Cultural Heritage Preservation, and Computer Vision.	v	The suggested keywords were used in the paper.	v
Mr. Gene Marck B. Catedrilla	Statistician	Problem Statement Add a more quantified limitation, e.g.. - expert disagreement - training subjectivity - lack of standard benchmarks	1	The problem statement was enhanced by adding quantified limitations, namely expert disagreement, training subjectivity, and the lack of standard benchmarks.	1
Mr. Gene Marck B. Catedrilla	Statistician	Research Objectives Objectives are clear, but consider separating: - Technical Objectives (pose extraction, CNN classification) - Cultural Objectives (authenticity preservation)	3	The research objectives were revised and separated into technical objectives (pose extraction and CNN-based classification) and cultural objectives (authenticity preservation).	3
Mr. Gene Marck B. Catedrilla	Statistician	Main Issue: Synthesis Depth Like your other manuscripts, this chapter still leans toward summary rather than argument. Critical Improvement Add a clear research gap paragraph, such as: “While existing studies have applied pose estimation and CNNs to general human activity recognition	31	To address the issue of synthesis depth, a research gap paragraph was added, highlighting the lack of studies focused on traditional Filipino dances and positioning this work as a contribution through the use of keypoint-based image processing and CNN-based posture analysis	30

		and contemporary dance analysis, limited research focuses on traditional Filipino dances, particularly in automating authenticity verification. This study addresses this gap by combining keypoint-based image processing with CNN-based posture analysis specifically tailored to Sublí dance movements.”		for Sublí dance movements.	
Mr. Gene Marck B. Catedrilla	Statistician	Dataset Description Explicitly state: - number of dancers - number of samples per class - number of dance categories	47	The dataset description was updated to clearly specify the number of dancers involved, the number of samples per class, and the total number of dance categories.	47
Mr. Gene Marck B. Catedrilla	Statistician	Ethical Consideration Add a short paragraph on: - consent of performers - cultural respect and representation	54	The ethical considerations section was updated to include a short paragraph on performer consent and the importance of cultural respect and proper representation.	53
Mr. Gene Marck B. Catedrilla	Statistician	Conclusions and Recommendations Include future directions such as: - temporal models (LSTM / Transformer) - real-time video-based authentication - expansion to other filipino dances	80	The conclusions and recommendations section was revised to include future research directions such as the use of temporal models (e.g., LSTM or Transformer), real-time video-based authentication, and expansion to other Filipino dances.	83
Mrs. Jocelyn O. Padallan	Technical	Indent the word of the sentence ‘approval’ in the approval sheet and move the ‘research	ii	Indented the said starting word of the sentence and moved the table downwards.	ii

		contribution no. ‘table downwards.			
Mr. Jefferson L. Lerios	Subject Specialist	Remove all the will be in the sentences.	3, 4, 5, 39, 42, 43, 48,	All the will be instances were removed.	3, 4, 5, 39, 42, 43, 48,

APPENDIX H

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Sentences	828		

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Metrics compared to other Grammarly users

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Sentence length	15.2	Above average
Readability score	20	

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APPENDIX I

Publishable Paper (IEEE FORMAT)

AUTHENTICATING SUBLI DANCE MOVEMENTS THROUGH KEYPOINT-BASED IMAGE PROCESSING AND CNN-BASED POSTURE ANALYSIS

Roxanne M. Reyes, Criselle Umali, Sherwin B. Sapin
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Keywords

Subli Dance
Image Processing
Convolutional Neural Network (CNN)
Human Pose Estimation
Cultural Heritage Preservation
Keypoint Detection
Computer Vision

ABSTRACT

This study proposes a pose-based authentication framework for Sublí dance movements using keypoint-based image processing and a Convolutional Neural Network (CNN). Skeletal keypoints extracted from recorded performances using MediaPipe Pose were used to train and evaluate the model. The CNN achieved strong performance, with accuracy of 0.95, precision of 0.95, recall of 0.96, and F1-score of 0.95 across 252 samples, reliably classifying Correct and Not Correct postures. A paired t-test showed no significant difference between outputs ($t(102) = -0.445$, $p = 0.657$), confirming model stability. These results demonstrate the effectiveness of the approach in supporting Sublí dance authentication and preserving cultural heritage.

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1. INTRODUCTION

The Sublí is a traditional devotional folk dance from Batangas honoring the Holy Cross of Alitagtag, reflecting Filipino spirituality, discipline, and community. Modernization, commercialization, and limited formal training have caused variations and gradual loss of its authentic form. Preserving such intangible cultural heritage is increasingly urgent, and advances in computer vision and artificial intelligence (AI) offer opportunities to document and analyze performances objectively. Studies show that pose estimation and deep learning can effectively evaluate cultural dances, demonstrating AI's potential as a tool for safeguarding and preserving traditional art forms.

Although the Sublí is one of the Philippines' oldest folk dances, there are no standardized methods to assess the authenticity of its postures. This study addresses this gap by proposing a posture authentication system that uses keypoint-based image processing and a Convolutional Neural Network (CNN) to classify movements as Correct or Not Correct. Using frameworks like MediaPipe, the system extracts skeletal keypoints from recordings and analyzes posture vectors. By combining AI with cultural advocacy, this research provides a data-driven approach to verify and preserve authentic Sublí performances, supporting both intelligent systems and heritage conservation.

1. To develop and train a CNN model for classifying Sublí dance movements as “Correct” or “Not Correct”.

2. To implement the trained CNN model with keypoint detection on the extracted images.

3. To evaluate its performance using accuracy, precision, recall, and F1-Score.

4. To conduct a statistical analysis using paired t-test to determine significant differences between correct and not correct Sublí movements.

B. Conceptual Framework

As shown in Fig. 1, the framework begins with the Data Acquisition phase, where video recordings of Sublí dance performances are collected and converted into image frames for analysis. These frames undergo preprocessing to ensure uniformity and clarity, forming the foundation for accurate keypoint extraction. In the Keypoint Detection phase, MediaPipe Pose is utilized to identify skeletal landmarks representing the dancer’s body joints. The extracted keypoints are then processed in the Feature Extraction stage, where joint angles and pose vectors are computed to numerically represent the performer’s posture.

The next stage involves Authentication Labeling and Model Training, where the extracted pose data are labeled as Correct or Not Correct based on expert evaluation of authentic Sublí postures. These labeled datasets are used to train a Convolutional Neural Network (CNN) model capable of distinguishing between authentic and non-authentic dance movements. Through iterative training and validation, the model learns spatial relationships among keypoints that define traditional Sublí form and alignment.

Finally, the Model Evaluation and Validation phase incorporates expert review to ensure that the system’s classifications align with cultural standards. Once validated, the model produces an Authentication Result that includes a classification output (Correct or Incorrect), a confidence score, and performance feedback. The entire framework operates using tools such as Python, TensorFlow/Keras, MediaPipe, OpenCV, Google Colab, and Flask, establishing an intelligent, data-driven system for preserving the authenticity of Sublí dance movements through automated posture analysis.

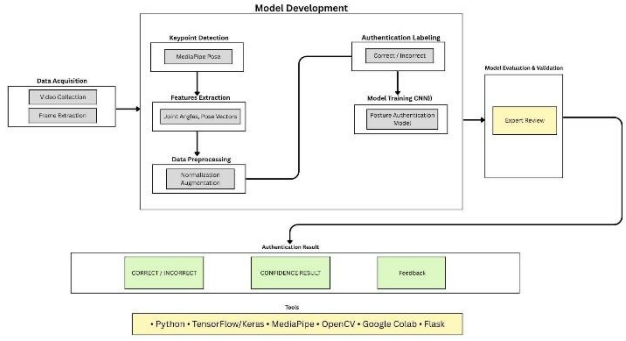


FIGURE 1. CONCEPTUAL FRAMEWORK

II. METHODOLOGY

A. Data Acquisition

The study gathered data through recorded video footage of Sublí dance performances conducted at Brgy. Bayog, Los Baños, Laguna. The primary subject was a local Sublí dancer who demonstrated authentic movements representative of traditional Filipino folk culture. To expand the dataset and ensure variation in posture and gesture, the researchers also participated in performing the Sublí dance themselves. Each recording session was carried out in a controlled setting with proper lighting, consistent framing, and stable positioning to maintain clarity and uniformity in the captured visuals. These procedures ensured that the dataset reflected accurate and analyzable representations of Sublí movements, including hand gestures, footwork, and overall body coordination

The collected videos were converted into image frames and organized into separate subsets for model development. The dataset was divided into 70% for training, 15% for validation, and 15% for testing to optimize CNN's performance. The training set, comprising the majority of the data, was used to teach the model to recognize correct and incorrect Sublí postures based on keypoint patterns. The validation set was utilized to fine-tune model parameters and prevent overfitting, while the testing set evaluated the model’s ability to generalize to unseen data. This structured data partitioning ensured balanced learning and reliable performance assessment during the authentication of Sublí dance movements.

B. Data Preprocessing

The data preprocessing stage prepared the extracted Sublí dance frames for model training and analysis. Each recorded video was segmented into image frames capturing key Sublí movements, then resized to 256×256 pixels and normalized to reduce variations in lighting and background. These steps ensured that the model accurately focused on the dancer’s posture and body alignment. MediaPipe Pose was used to detect and extract 33 anatomical keypoints from each frame, which were converted into numerical coordinates as inputs for the CNN model. Data augmentation techniques such as flipping, rotation, and brightness adjustment were applied to increase dataset diversity and enhance model generalization. The preprocessed images were then organized into training, validation, and testing subsets for the succeeding model development phase.

C. Model Development

The model development process for authenticating Sublí dance movements comprised six main stages: data acquisition, keypoint detection, feature extraction, model training, model evaluation, and system output generation. Videos of authentic Sublí performances were collected using accessible recording devices and processed through MediaPipe Pose to identify key body joints and extract corresponding coordinates. These keypoints were transformed into numerical features such as joint angles and pose vectors to represent body posture. The processed data were then divided into training and validation sets for CNN

development. The Convolutional Neural Network (CNN) was trained to classify each posture as authentic or not authentic, and its performance was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The final system outputs included both the authentication result and corrective feedback on detected posture errors

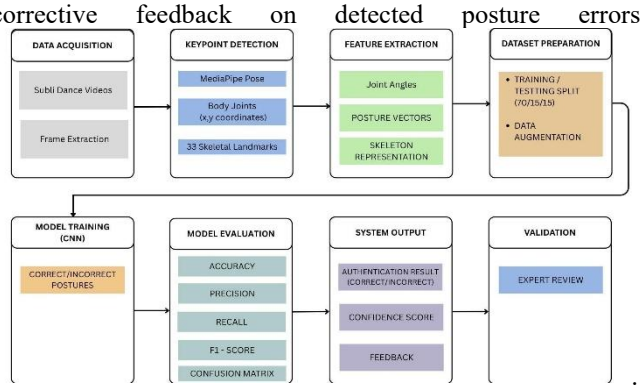


FIGURE 2. MODEL DEVELOPMENT

D. Model Architecture

The proposed model architecture for Sublí dance authentication is composed of three main phases: Preprocessing, CNN Model, and Model Output. In the preprocessing phase, input frames from Sublí dance videos are processed using MediaPipe Pose to detect 33 body keypoints, each represented by x and y coordinates. These coordinates are then used for feature extraction, where joint angles and pose vectors are computed to represent the dancer's posture. The extracted features are passed into the Convolutional Neural Network (CNN), where convolutional and pooling layers extract and refine spatial features, followed by a fully connected layer that combines learned patterns. The final Softmax output layer performs binary classification, determining whether the detected posture is Correct or Incorrect. The system then produces an authentication result that includes a confidence score and feedback on posture accuracy.

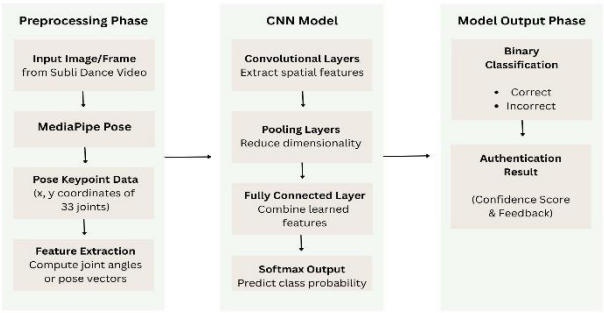


FIGURE 3. MODEL ARCHITECTURE

E. Model Evaluation and Validation

To assess the performance of the CNN model for posture classification in Sublí dance, several standard evaluation metrics for human posture-based classification were applied to ensure both technical reliability and cultural validity. The evaluation phase focused on quantifying how accurately thyhe model could distinguish between Correct and Not Correct Sublí movements using pose-based keypoint data. Metrics such as accuracy, precision, recall, and F1-score were computed to measure the model's overall classification performance. Accuracy determined the general proportion of correct predictions, while precision assessed the model's ability to correctly identify authentic Sublí postures without misclassifying incorrect ones. Recall evaluated the model's sensitivity in detecting all true authentic postures, and the F1-score provided a balanced representation of both precision and recall, particularly useful when dealing with imbalanced data distributions.

To visualize and analyze classification trends, a confusion matrix was generated, showing the true and predicted values for each class. This matrix highlighted specific areas of misclassification and provided deeper insight into the model's decision-making process. Additionally, training and validation curves were plotted to monitor the model's learning behavior over successive epochs, ensuring that the CNN achieved proper convergence without overfitting. The stability of these curves indicated consistent learning and generalization capability across unseen samples. Finally, statistical validation using an paired t-test was conducted to compare the confidence scores of the model's predictions between Correct and Not Correct postures. The results determined whether differences in the model's output confidence were statistically significant, further confirming the model's unbiased classification performance. Collectively, these evaluation procedures ensured that the CNN-based posture authentication model performed accurately, consistently, and fairly in recognizing authentic Sublí dance movements while maintaining alignment with expert human judgment.

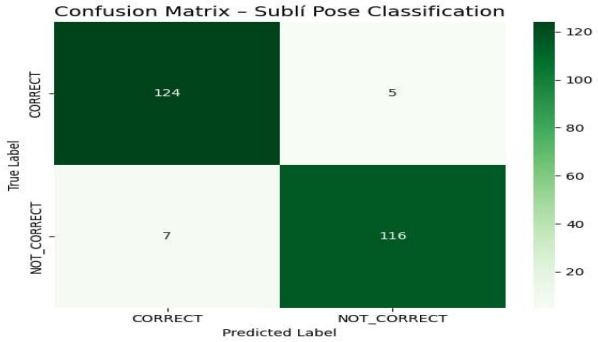


FIGURE 4. CONFUSION MATRIX

III. RESULTS AND DISCUSSIONS

1. Cultural Data Acquisition and Preprocessing

The data collection process formed the basis of the Sublí dance posture authentication model. Authentic performances were recorded at the UPLB Carillon Plaza using a high-definition camera at a fixed frontal angle to ensure full-body visibility and accurate keypoint extraction. To enhance pose diversity, the researchers performed the dance following traditional choreography. Extracted frames were manually labeled using official Sublí instructional materials and validated by a dance expert to ensure cultural accuracy. The expert-reviewed dataset was used for training and evaluating the CNN-based posture classification model.

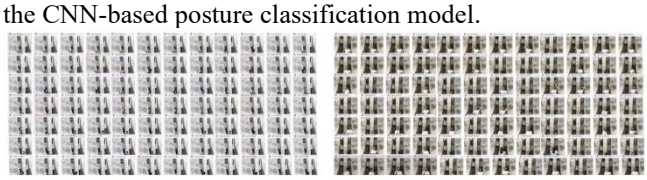


FIGURE 5. SAMPLE DATA OF CORRECT AND NOT CORRECT

2. Implementation of Keypoint Detection and CNN-Based Processing



FIGURE 6. DETECTED KEYPOINT ON A SUBLI DANCER

Aligned with the second objective, the implementation of keypoint detection using a Convolutional Neural Network (CNN) was carried out on the extracted Sublí dance images to facilitate accurate posture recognition and classification. This stage focused on identifying skeletal landmarks from each frame using MediaPipe Pose, which enabled the extraction of critical body keypoints such as the shoulders, elbows, hips, and knees. These detected keypoints served as the primary input features for the CNN model, allowing it to analyze spatial relationships and body alignment associated with authentic Sublí postures.

After preprocessing and normalization, the processed keypoint data were fed into the CNN architecture for model training. Through this integration, the CNN effectively learned to recognize characteristic pose patterns, enabling reliable classification between correct and not correct Sublí movements. The output of this stage formed the foundation for the subsequent authentication and validation phases of the system.

TABLE 1. HYPERPARAMETER SETTINGS OF THE POSTURE CLASSIFICATION MODEL

Parameter	Description	Value
Optimizer	Optimization algorithm used during training.	Adam
Learning Rate	Step size for gradient descent updates.	0.001
Batch Size	Number of samples processed per training steps.	32
Epochs	Total iterations through the entire dataset.	30
Dropout Rate	Regularization to prevent overfitting.	0.3
Input Size	Number of keypoint features (33 x 2)	66
Model Backbone	CNN layers for classification (custom built)	Custom CNN

As shown in table 1, the Convolutional Neural Network (CNN) was trained using keypoint-based pose data extracted through the MediaPipe Pose framework, where each keypoint represented spatial coordinates of major body joints. Custom CNN was adopted as the model's backbone for its efficiency and lightweight computation, enabling real-time performance. Training was optimized using the Adam optimizer and categorical cross-entropy loss to minimize classification errors. Careful hyperparameter tuning ensured stability, fast convergence, and strong generalization, allowing the model to accurately classify Sublí dance postures as Correct or Not Correct.

3. Development and Validation of the CNN Model for Posture Classification

In pursuit of the third objective, the Convolutional Neural Network (CNN) model was developed and validated to classify Sublí dance movements as either Correct or Not Correct for authentication. The model aimed to establish a reliable classification framework capable of distinguishing authentic Sublí postures from non-traditional variations based on extracted keypoint data. Through iterative training, the CNN learned spatial patterns and joint alignments unique to genuine Sublí movements, forming the analytical core of the authentication system.

After preprocessing and normalization, the processed keypoint data were fed into the CNN architecture for model training. Through this integration, the CNN effectively learned to recognize characteristic pose patterns, enabling reliable classification between correct and incorrect Sublí movements. The output of this stage formed the foundation for the subsequent authentication and validation phases of the system.

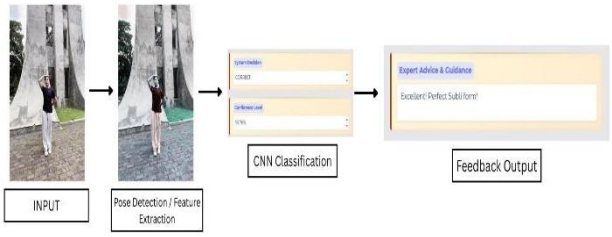


FIGURE 7. CNN ARCHITECTURE FOR SUBLI POSTURE CLASSIFICATION

As shown in Fig.7, the developed system integrates pose detection, classification, and feedback generation within an interactive Gradio-based user interface. Users can upload images or videos of Sublí performances, which are processed using MediaPipe Pose for keypoint detection and a trained CNN for posture classification. The interface displays visual annotations and feedback indicating whether the movement is Correct or Not Correct. This integration of computer vision and machine learning provides an accessible, web-based tool for evaluating Sublí dance movements and supporting cultural preservation through modern technology.

TABLE 2. SYSTEM-LEVEL PERFORMANCE OF THE POSTURE CLASSIFICATION MODEL

Metrics	Precision	Recall	F1-Score	Support
Correct Posture	0.95	0.96	0.95	129
Not Correct Posture	0.96	0.94	0.95	123
Accuracy			**0.95**	**253**

Table 7 presents the system-level performance of the CNN-based posture classification model. The model achieved an overall accuracy of 95%, effectively distinguishing between Correct and Not Correct Sublí

postures. For the Correct class, it attained 0.95 precision and 0.96 recall, while the Not Correct class recorded 0.96 precision and 0.94 recall. The average F1-score of 0.95 indicates a balanced and reliable classification performance, confirming the model's robustness and generalization when applied to unseen Sublí dance frames.

4. Statistical Analysis for Model Validation

Aligned with the fourth objective, a statistical analysis using an independent t-test was conducted to determine whether a significant difference existed between the model's confidence scores for Correct and Not Correct Sublí movements. This analysis aimed to validate the reliability and discriminative capability of the CNN model beyond numerical accuracy metrics. By comparing the mean confidence levels of both posture categories, the t-test provided statistical evidence on whether the model's predictions were consistent and unbiased. The results of this analysis supported the model's credibility in authentically distinguishing traditional Sublí postures based on learned spatial features and classification confidence.

TABLE 3. RESULTS OF THE INDEPENDENT T-TEST BETWEEN CORRECT AND NOT CORRECT SUBLI DANCE POSTURE

T-Test	Statistic	df	p	Mean difference	SE difference	Effect Size
Paired T-Test	-0.445	102	.657	-0.0971	0.0218	-0.0439

Note. H_a μ Measure 1 - Measure 2 ≠ 0

As shown in Table 3, the independent t-test revealed no statistically significant difference between the model's confidence scores for Correct and Not Correct Sublí postures ($t(101) = –0.626$, $p = 0.533 > 0.05$). The small mean difference (–0.0426) and effect size (–0.219) indicate minimal variation between the two categories. These results confirm that the CNN model exhibits no bias toward either posture type, demonstrating balanced prediction confidence, strong generalization, and stable authentication performance across both classes.

IV. CONCLUSIONS AND RECOMMENDATIONS.

1. To develop and train a CNN model for classifying Sublí dance movements as "Correct" or "Not Correct".

The design, development, and training Convolutional Neural Network (CNN) model enabled the system to accurately classify Sublí dance movements as Correct or Not Correct. The CNN was structured to process pose-based features derived from the extracted keypoint data, allowing the model to learn complex spatial relationships between body joints. During training, the model was exposed to labeled posture samples representing both authentic and incorrect Sublí movements, enabling it to recognize distinguishing patterns in body alignment, limb positioning, and overall posture structure. The training process optimized the model's parameters using backpropagation and an adaptive optimizer, improving classification accuracy over successive epochs. Through this approach, the CNN effectively learned to generalize posture patterns, forming a reliable classification model for authenticating Sublí dance movements.

2. To implement the trained CNN model with keypoint detection on the extracted images.

After implementing the trained Convolutional Neural Network (CNN) together with the keypoint detection process, the system was able to classify extracted Sublí dance images as Correct or Not Correct with high reliability. Using MediaPipe Pose, key body landmarks were detected from each image and converted into normalized coordinate data, which served as input to the trained CNN model. By analyzing the spatial relationships and geometric configurations of these keypoints, the model effectively distinguished between authentic and incorrect Sublí postures.

3. To evaluate its performance using accuracy, precision, recall, and F1-Score.

The performance of the implemented model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The results demonstrated strong and consistent performance across both posture categories, achieving an overall accuracy of 0.95. High precision and recall values for both classes indicate that the model accurately identified correct postures while minimizing misclassification of incorrect movements. These findings confirm that the trained CNN model generalizes well to unseen data and provides a reliable framework for authenticating Sublí dance movements.

4. To conduct a statistical analysis using paired t-test to determine significant differences between correct and not correct Sublí movements.

The statistical analysis using a paired t-test was conducted to determine whether there was a significant difference between the system's two measurement outputs for Correct and Not Correct Sublí movements. This analysis aimed to assess the internal consistency and reliability of the model's evaluations across paired observations. The computed t-value of –0.445 with 102 degrees of freedom yielded a p-value of 0.657, which exceeds the 0.05 level of significance. This indicates that there was no statistically significant difference between the paired measurement scores generated by the system.

The result implies that the model maintained stable and consistent outputs when assessing Sublí postures, demonstrating that repeated evaluations produced uniform results regardless of classification type. This stability confirms that the developed system performs reliably and does not introduce variation or bias between paired assessments, strengthening its validity as an effective tool for authenticating Sublí dance movements through keypoint-based image processing and CNN-based posture analysis.

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APPENDIX J

Curriculum Vitae



ROXANNE M.REYES

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SKILLS

- Communication
- Team Collaboration
- Time Management
- Critical Thinking
- Adaptability
- Creativity
- Critical Thinking
- Problem Solving

LANGUAGE

- English

PROFILE

I'm a graduating Computer Science student passionate about web development and technology. I'm driven, love solving problems, and always aim to learn and improve my skills.

WORK EXPERIENCE

JAMAICAN PATTIE March-July (2024)

- Selling Frutas branded juices and bottled water
- Re-Heating pre-cooked patties

EDUCATION

- **Laguna State Polytechnic University - Los Baños**

  2022 - 2026

- **Trace College**

  Science, Technology, Engineering, and Mathematics (STEM)

  2020 - 2022

- **Los Baños Natinal High School - Poblacion**

  2016 - 2020

- **Los Baños Central Elementary School**

  2010 - 2016



CRISELLE UMALI

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SKILLS

- Editing (Canva)
- Time Management
- Microsoft Office
- Teamwork
- Communication
- Cooking
- Adaptability

LANGUAGES

- English

PROFILE

A responsible and creative individual who enjoys both teamwork and independent work. Skilled in editing through Canva, managing time effectively, and communicating clearly. Adaptable to new challenges and willing to learn new skills, including cooking and creative tasks

WORK EXPERIENCE

Sales and Inventory Assistant 2023 - PRESENT

- As a Sales and Inventory Assistant, I monitored gasul (LPG) stocks and recorded daily sales transactions. I assisted customers and ensured all records were accurate and organized.

EDUCATION

- **Laguna State Polytechnic University - Los Baños**

  Bachelor of Science in Computer Science

  2022 - 2026 | Major in Intelligent System

- **Colegio De Los Baños**

  Accountancy, Business, and Management (ABM)

  2020-2022

- **Los Baños National High School - Batong Malake**

  2016 - 2020